

Entity Relationship Diagrams vs UML Class Diagrams to model databases

A survey, original experiment and
replications

Carmine Gravino

Outline

- The beginning
- The three chapters
 - The design of the experiments/replications (3+4+5)
 - Results
- Summary and Discussion

The Beginning

- **Data models**
 - describe *the structure of (and the operations on) a database* in a diagrammatic form aiming at the communication between the stakeholders involved in the data modeling process
- The comprehensibility of such models is vital to avoid misunderstanding during the requirement elicitation phase as well as the implementation and modification of the database
 - ...
 - users vs database professionals
 - database modelers vs developers/mantainers

The crucial role of notation/graphical representation

The Beginning (cont.)

- The use of (E)ER in the Databases courses

still remains the *de facto* standard for DB conceptual modeling

- The use of UML in the Software Engineering courses

becoming the *de facto* standard for the analysis and design of software systems

- Two notation employed for data modeling during software development process!

Why?

- Comparing ER and UML class diagrams
- In a small industrial survey we found that c.a. 50% software companies usually employ both ER and UML class diagrams to represent the same database

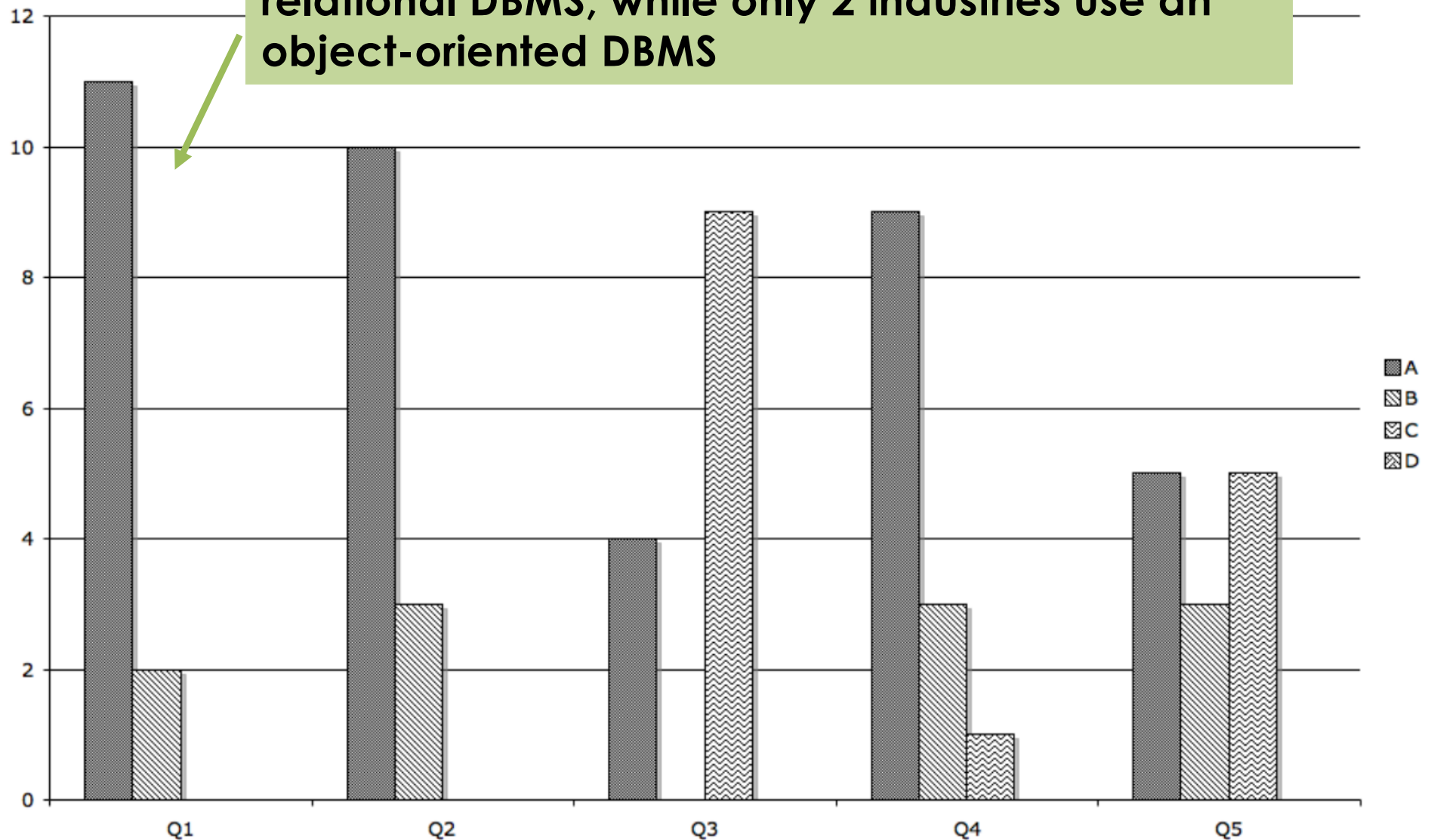
A Survey

- A questionnaire (by email) to 50 industrial partners and 13 of them filled-in the questionnaire

Id	Question
Q1	In general, how do you manage the persistent data? (A) R-DMBS (B) OO-DBMS (C) Other
Q2	Which data model notation do you use in your company? (A) ER (B) UML (C) Other
Q3	Did you ever use different notations for modeling the same database? (A) Yes (ER and UML) (B) Yes (Others) (C) No
Q4	According to your experience, which is the notation simpler to maintain? (A) ER (B) UML (C) Both (ER and UML) (D) Other
Q5	According to your experience, which should be the notation to teach in university courses? (A) ER (B) UML (C) Both (ER and UML) (D) Other

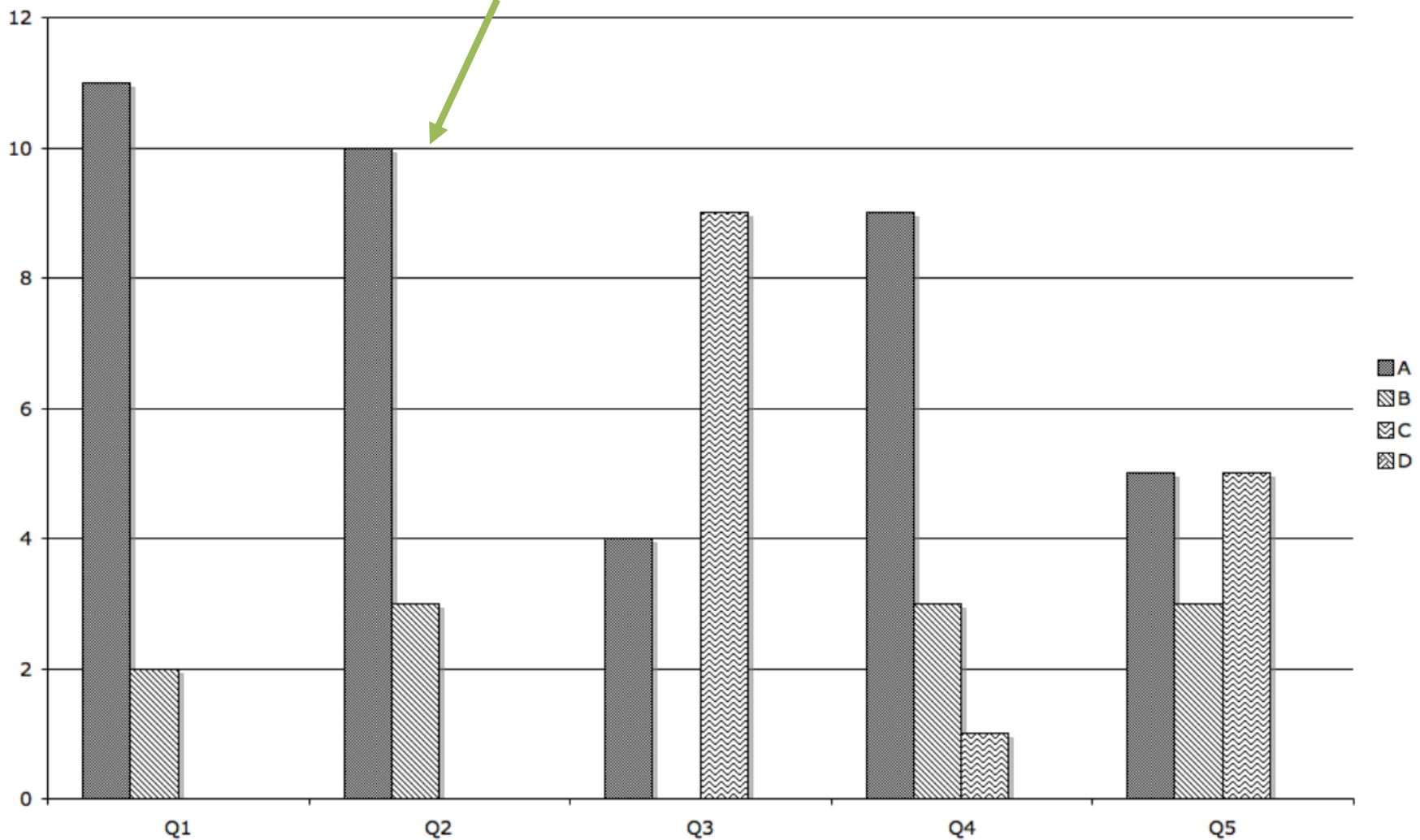
Survey results (cont.)

the majority of the contacted industries use a relational DBMS, while only 2 industries use an object-oriented DBMS



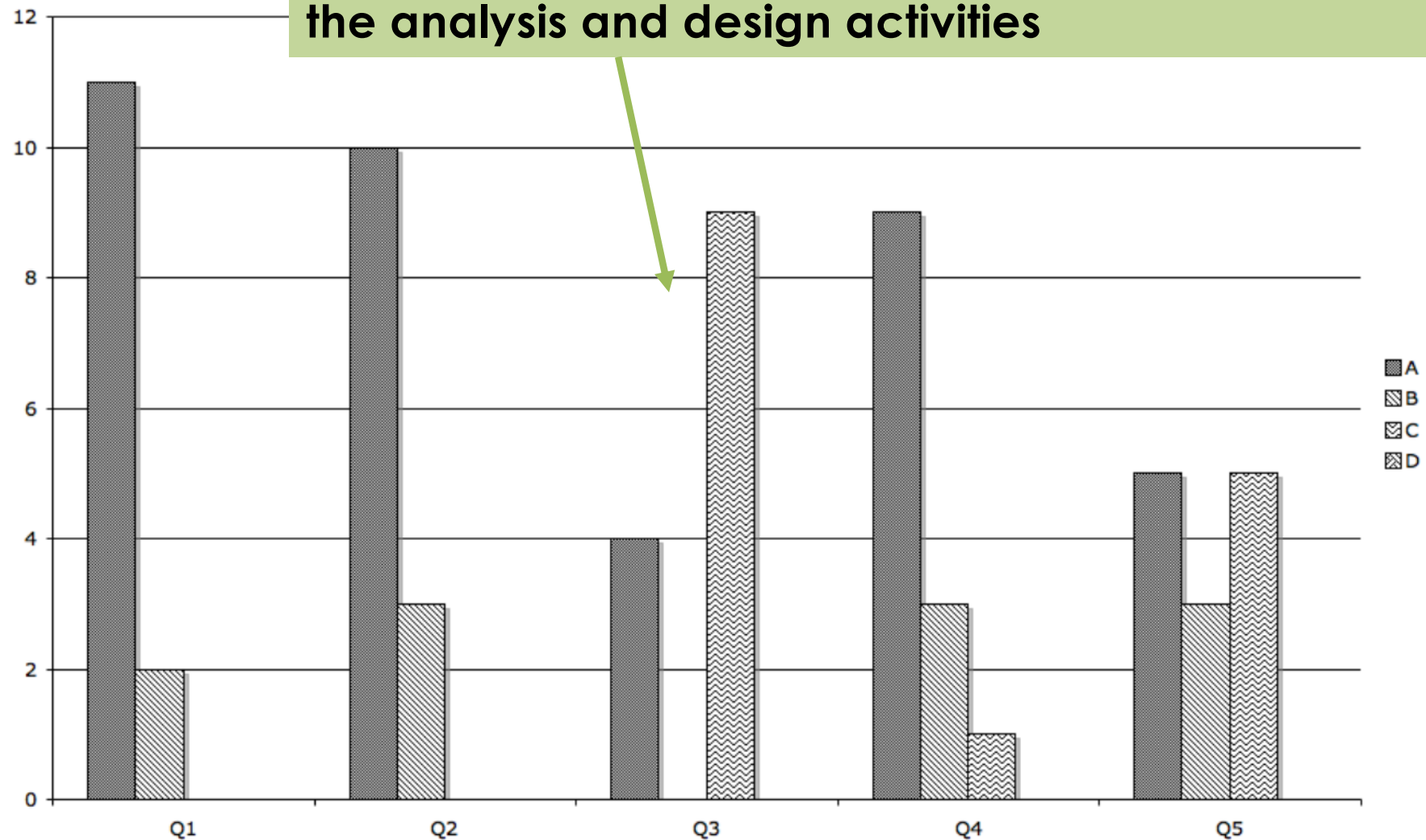
Survey results (cont.)

10 industries use ER diagrams, while only 3 use UML class diagrams



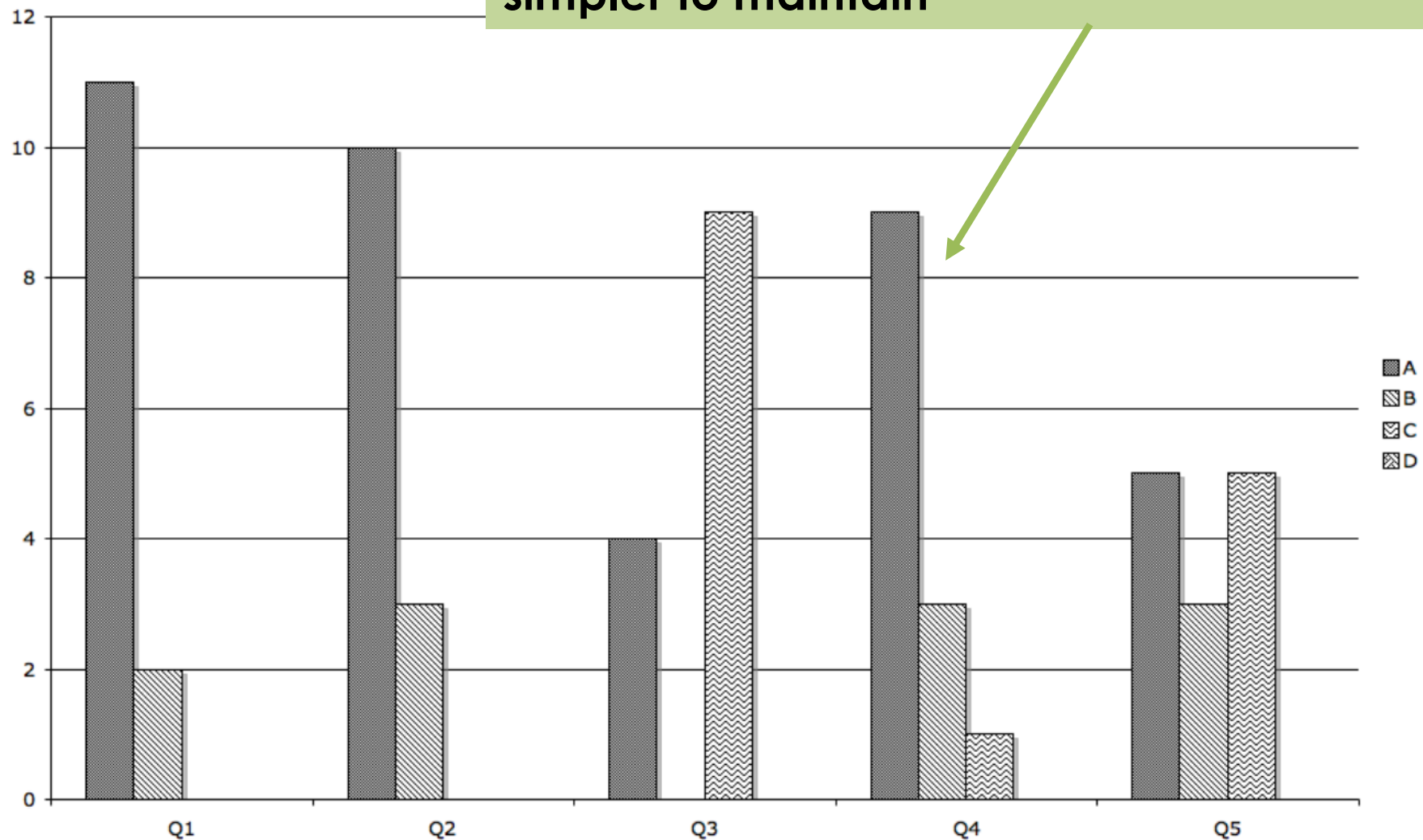
Survey results (cont.)

4 industries employ ER to model the database of the software system, even if they use the UML for the analysis and design activities



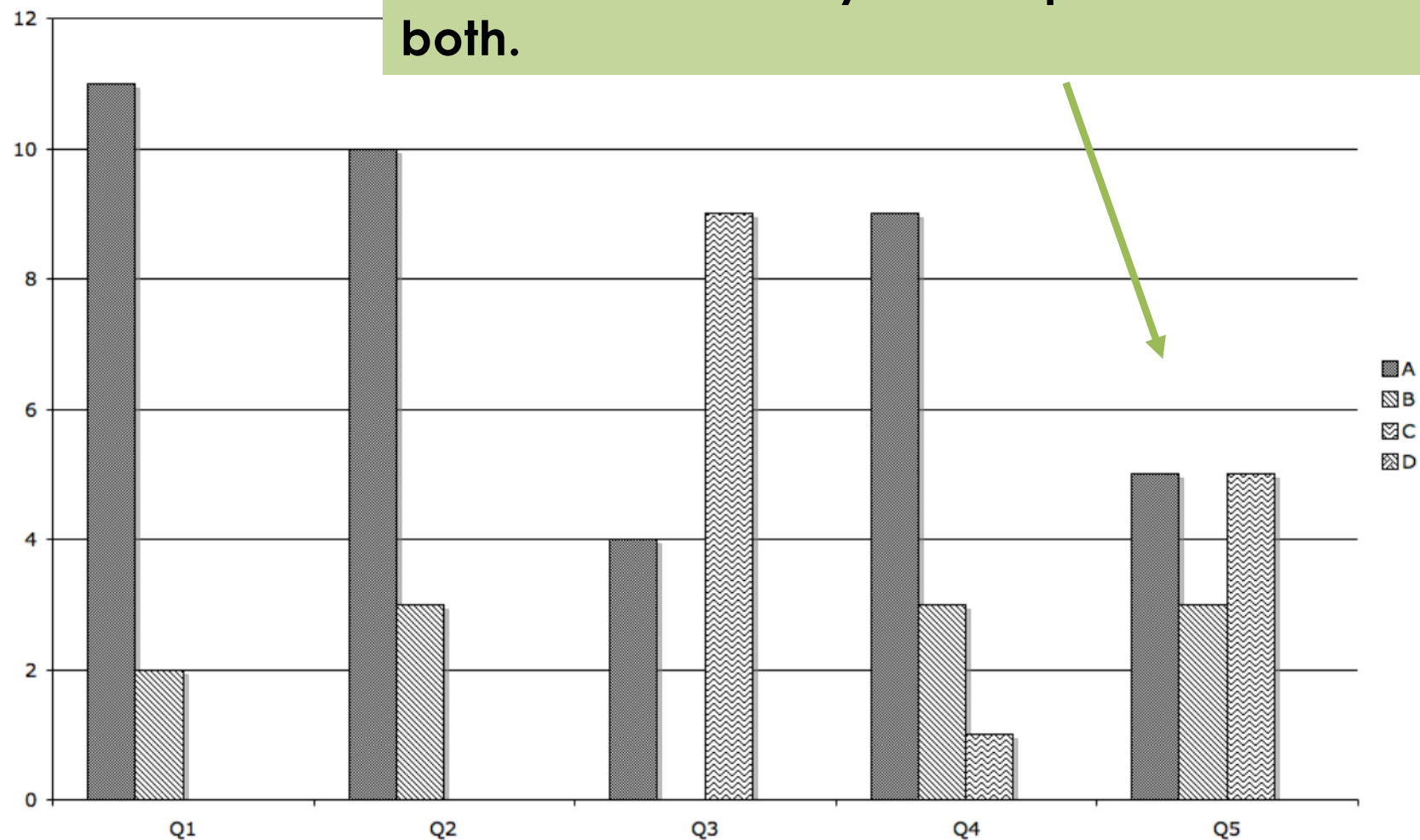
Survey results (cont.)

**9 found ER diagrams simpler to maintain,
while 3 found the UML class diagrams
simpler to maintain**

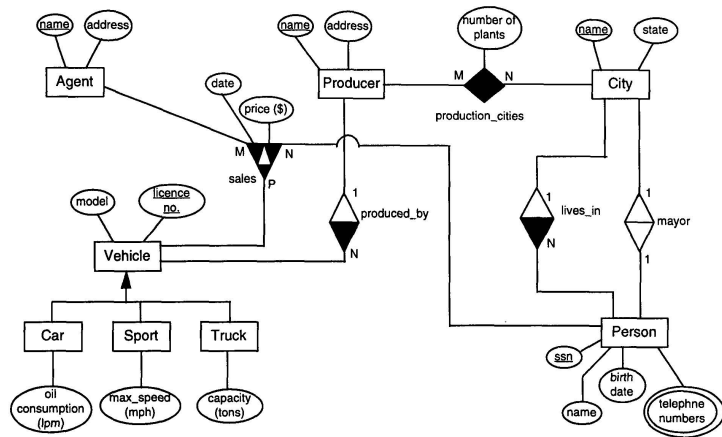


Survey results (cont.)

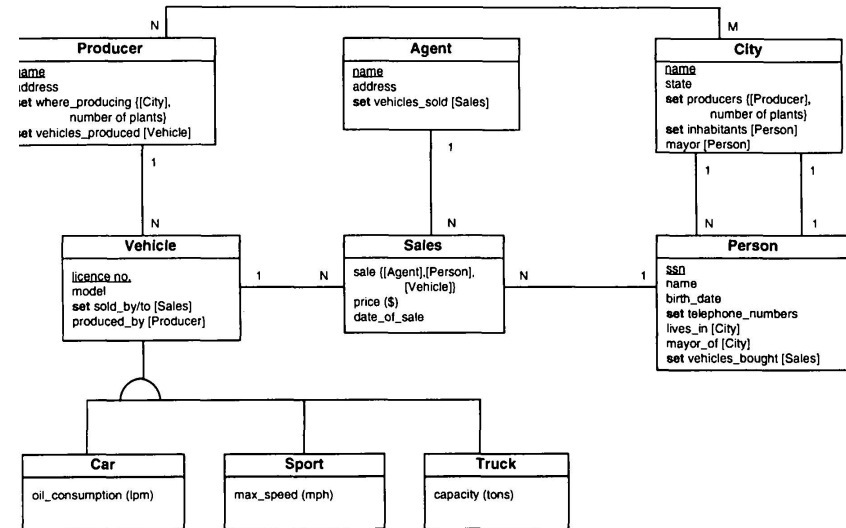
5 believe that is better to teach only the ER in academic course, while 3 believe that it is better to teach only UML. 5 partners believe both.



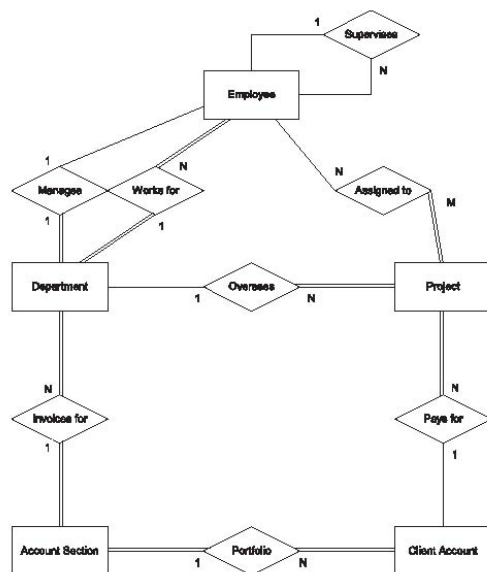
The literature



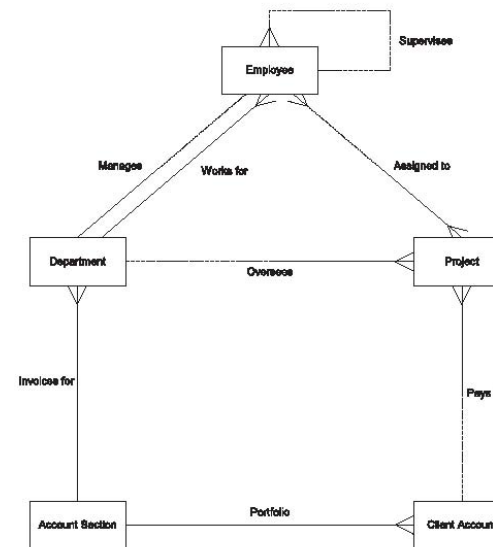
VS



Shoval et al - Entity-relationship and object-oriented data modeling
experimental comparison of design quality - Data & Knowledge Engineering 21 (1997) 297-315



VS



Purchase et al. - Comprehension of diagram syntax: an empirical study of entity relationship
Notations - Int. J. Human-Computer Studies 61 (2004) 187-203

Controlled Experiments?



The Beginning (cont.)

The chapters :

1. An experimental comparison of ER and UML class diagrams for data modeling (2007-2008)

2. A Fine-Grained Analysis of the Support Provided by UML Class Diagrams and ER Diagrams During Data Model Maintenance (2009-2010)

Considering single building blocks of ER and UML class diagrams

3. Using Stereotypes in UML-Based Data Modeling to overcome the Weaknesses of UML Class Diagrams (2011-2012)

Considering single building blocks of ER and UML class diagrams

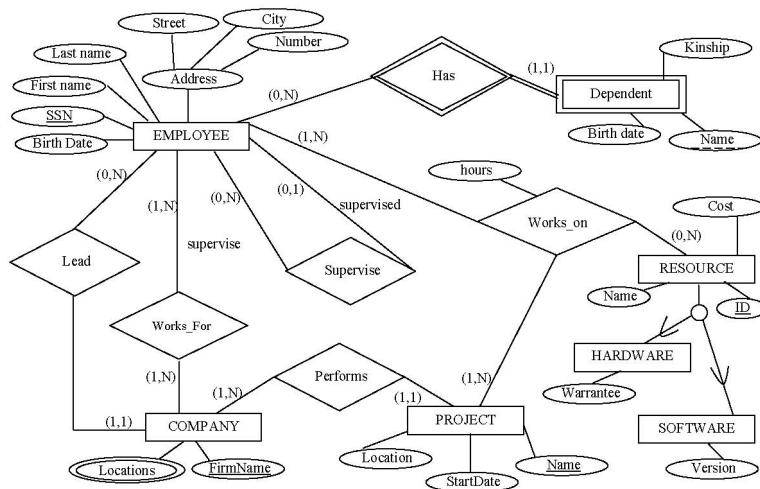
The Beginning (cont.)

The participants:

- Andrea De Lucia, Carmine Gravino, Rocco Oliveto, Genny Tortora (Chapter 1)
- Andrea De Lucia, Carmine Gravino, Rocco Oliveto, Genny Tortora; Marcela Genero, Jose Cruz Lemus
- (Chapter 2 and Chapter 3)
- Volunteers 489 (= 138 + 184 + 167)
(Students from University of Salerno and University of Castilla, La Mancha)
- and colleagues, editors, and anonymous reviewers

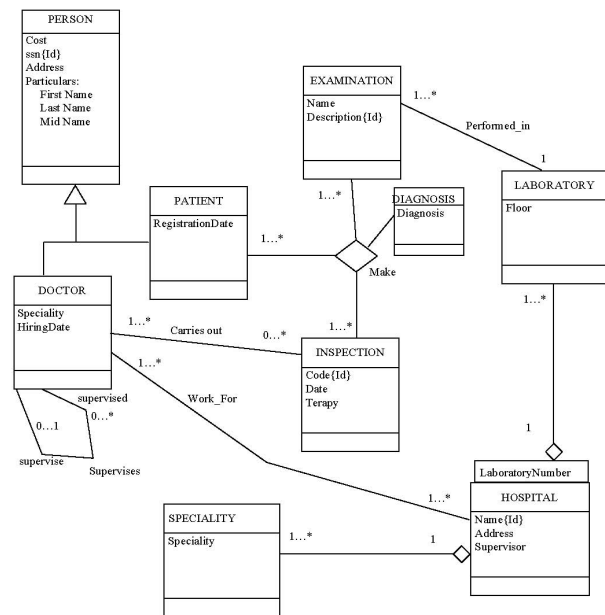
The Three Chapters: Chapter 1

An experimental comparison of ER and UML class diagrams for data modeling (2007-2008)



ER diagrams

VS



UML Class Diagrams

Design of experiments/replications

Goal Definition and Context

Goal

To analyse whether UML CDs are more comprehensible than ER diagrams during the comprehension and modification of data models

Systems

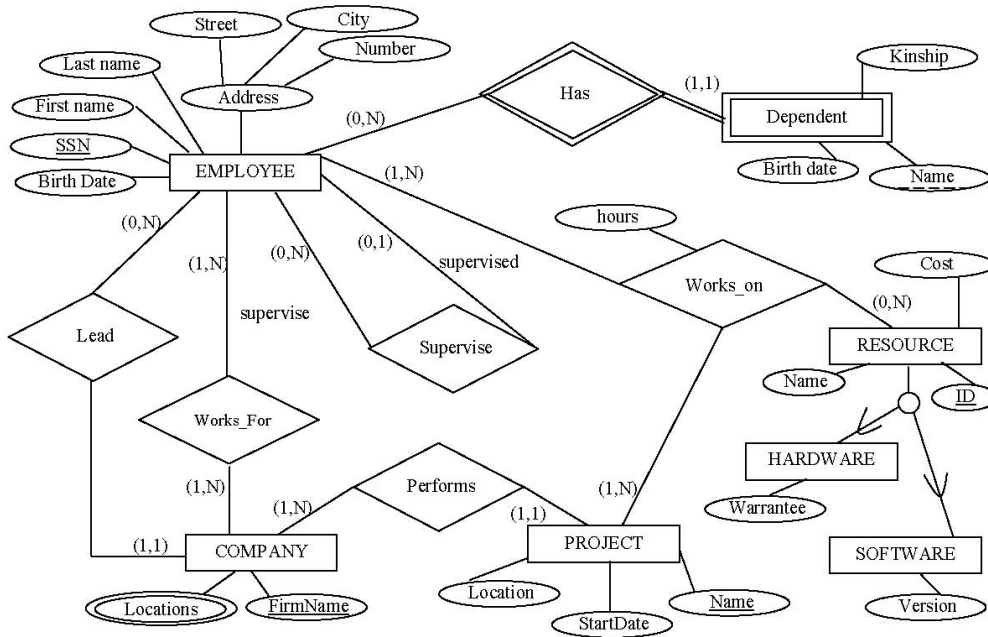
Data models from four systems:
i) ADAMS, ii) Company management system,
iii) EasyClinic, iv) University software system

Students from the University of Salerno with different background:

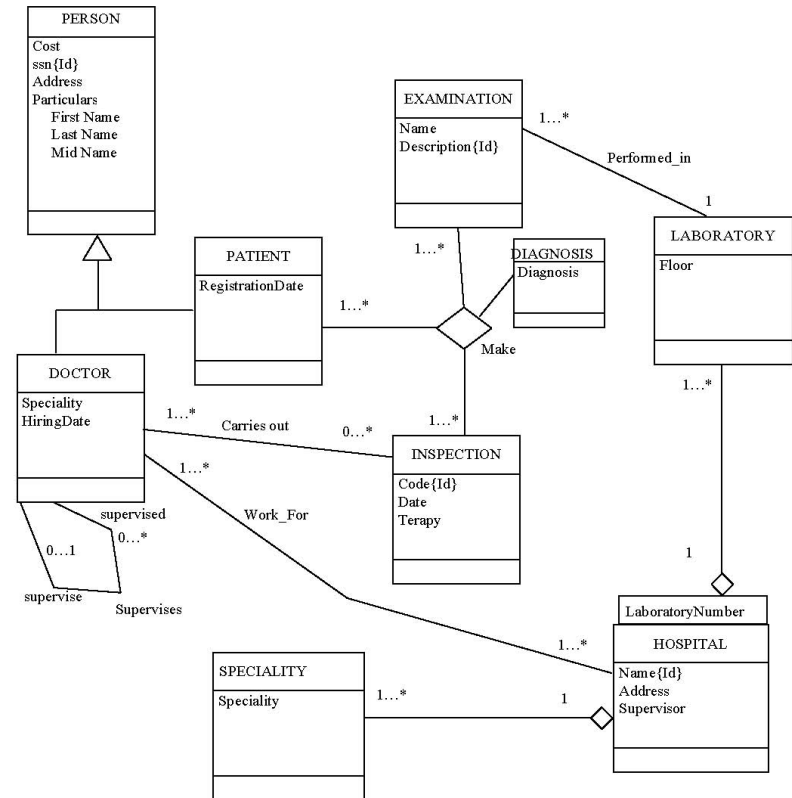
- bachelor students, 2nd year B.Sc. (n=40 in experiment, -both)
- master students, 1st year M.Sc. (n=30 in 1st replication, -both)
- fresher students, 1st year B.Sc. (n=68 in 2nd replication, -comprehension)

Design of experiments/replications (Cont.)

Systems



Company



EasyClinic

Miller's rule—a subdiagram with only 7 ± 2 entities of the original data models

Design of experiments/replications (Cont.)

Variable Selection

Independent variable

Method, i.e., the notation used (ER or UML CD)

Dependent variables

Comprehension level and **Modification level**, measured by the recall and precision measures

$$F - measure_s = 2 * \frac{precision_s * recall_s}{precision_s + recall_s} \%$$

$$recall_s = \frac{\sum_i |answer_{s,i} \cap correct_i|}{\sum_i |correct_i|} \% \quad precision_s = \frac{\sum_i |answer_{s,i} \cap correct_i|}{\sum_i |answer_{s,i}|} \%$$

Co-factors

- System
- Lab session

Other factors

- Experience
- Ability

Design of experiments/replications (Cont.)

Hypotheses

H_{0comp}: There is no difference in the **comprehension level** of participants when they have to comprehend **ER** or **UML CD**

H_{0Modi}: There is no difference in the **modification level** of participants when *identifying the change to perform on ER or UML CD to meet a maintenance request*
(we expect to reject the null hypotheses)

Experiment Design

Group	Treatment	
	ER	UML
A	S1, Lab1	S2, Lab2
B	S2, Lab2	S1, Lab1
C	S2, Lab1	S1, Lab2
D	S1, Lab2	S2, Lab1

Comprehension: S1 = ADAMS-TeamManagement, S2 = EasyClinic-BookingManagement

Maintenance: S1 = Company, S2 = University

Design of experiments/replications (Cont.)

Tasks

- **Questionnaire (Comprehension):**

- 10 multiple choice questions
- One or more correct answers

Q4 Let us focus on the classes Project and Company.

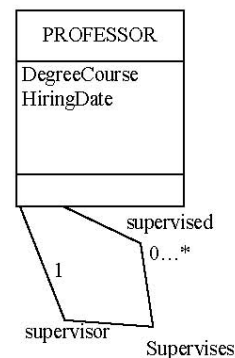
Which of the following statements is true:

- ☐ A company has a unique office
- ☐ A project has a unique office
- ☐ A company may have multiple offices

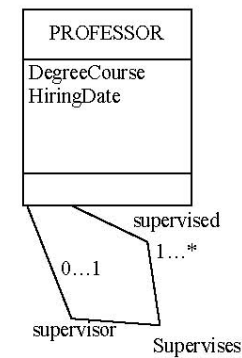
- **Questionnaire (Modification):**

- 5 multiple choice questions
- One correct answer

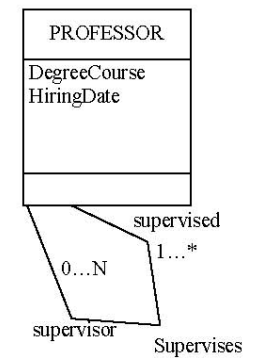
Q1 We want that each Professor is controlled by at least one other Professor. A Professor can control more than one Professor. Indicate the part of the class diagram interested by this modification activity:



(a)



(b)



(c)

Design of experiments/replications (Cont.)

Procedure and Data Analysis

- **Training** : on ER and UML CDs and on experiment execution.
- **Execution**: Tasks performed individually
- **Analysis**:

Behavior of the dependent variable

Descriptive statistics

Differences between treatments

Paired Wilcoxon test, each subject performed a task on two different models with the two possible treatments

Co-factors effect

ANOVA

Other factors effect

Interaction plots

Results

Comprehension level

Bachelor students

System	ER			UML		
	Median	Mean	Std. Dev.	Median	Mean	Std. Dev.
All	78.46	73.29	20.79	85.71	82.48	13.75
S1	75.96	70.79	20.95	85.71	82.65	11.29
S2	80.00	75.79	20.87	86.61	82.31	16.13

Master students

System	ER			UML		
	Median	Mean	Std. Dev.	Median	Mean	Std. Dev.
All	87.50	82.10	17.03	93.33	90.53	10.09
S1	87.50	82.21	19.02	90.42	86.68	11.15
S2	86.61	82.02	15.72	96.67	94.93	6.68

Fresher students

System	ER			UML		
	Median	Mean	Std. Dev.	Median	Mean	Std. Dev.
All	71.00	68.35	20.81	76.00	75.65	14.45
S1	71.00	67.24	19.42	73.00	75.00	14.63
S2	71.00	69.47	22.36	77.00	76.29	14.45

Results (cont.)

Modification level

Bachelor students

System	ER			UML		
	Median	Mean	Std. Dev.	Median	Mean	Std. Dev.
All	80.00	67.25	22.64	60.00	65.02	21.12
S1	80.00	64.50	21.64	60.00	62.22	19.11
S2	70.00	70.00	23.84	70.00	67.82	23.10

Master students

System	ER			UML		
	Median	Mean	Std. Dev.	Median	Mean	Std. Dev.
All	80.00	73.18	24.90	70.00	68.26	24.57
S1	70.00	67.14	25.55	60.00	66.46	23.45
S2	85.45	78.46	23.87	80.00	70.32	26.53

Results (cont)

Wilcoxon tests and descriptive statistics of differences

Comprehension level

Experiment	#subj.	#obs.	Med.	Mean	Std. Dev.	p-value	% of Pos. effect
Comp-Fsc	68	136	7.00	7.29	27.08	0.028	61.76
Comp-Bsc	40	80	7.95	9.19	25.00	0.015	65.00
Comp-Msc	30	60	2.92	8.43	18.76	0.021	70.00
All	138	276	6.50	8.09	24.73	2.42E-04	58.70

Modification level

Experiment	#subj.	#obj.	Median	Mean	Std. Dev.	p-value	% of Pos. effect
Maint-Bsc	40	80	0.00	-2.23	31.52	0.706	37.50
Maint-Msc	30	60	0.00	-4.92	38.12	0.731	26.67
All	70	140	0.00	-3.38	34.27	0.799	35.71

Results

- The achieved results demonstrated that using UML class diagrams participants achieved **better comprehension** levels
 - *Confirming our conjecture: “UML class diagrams are easier to understand than ER diagrams due to their more concise graphical notation”*
- With regard to the support given by the two notations during **modification** activities the results demonstrated that the two notations are **comparable**

Final questionnaire

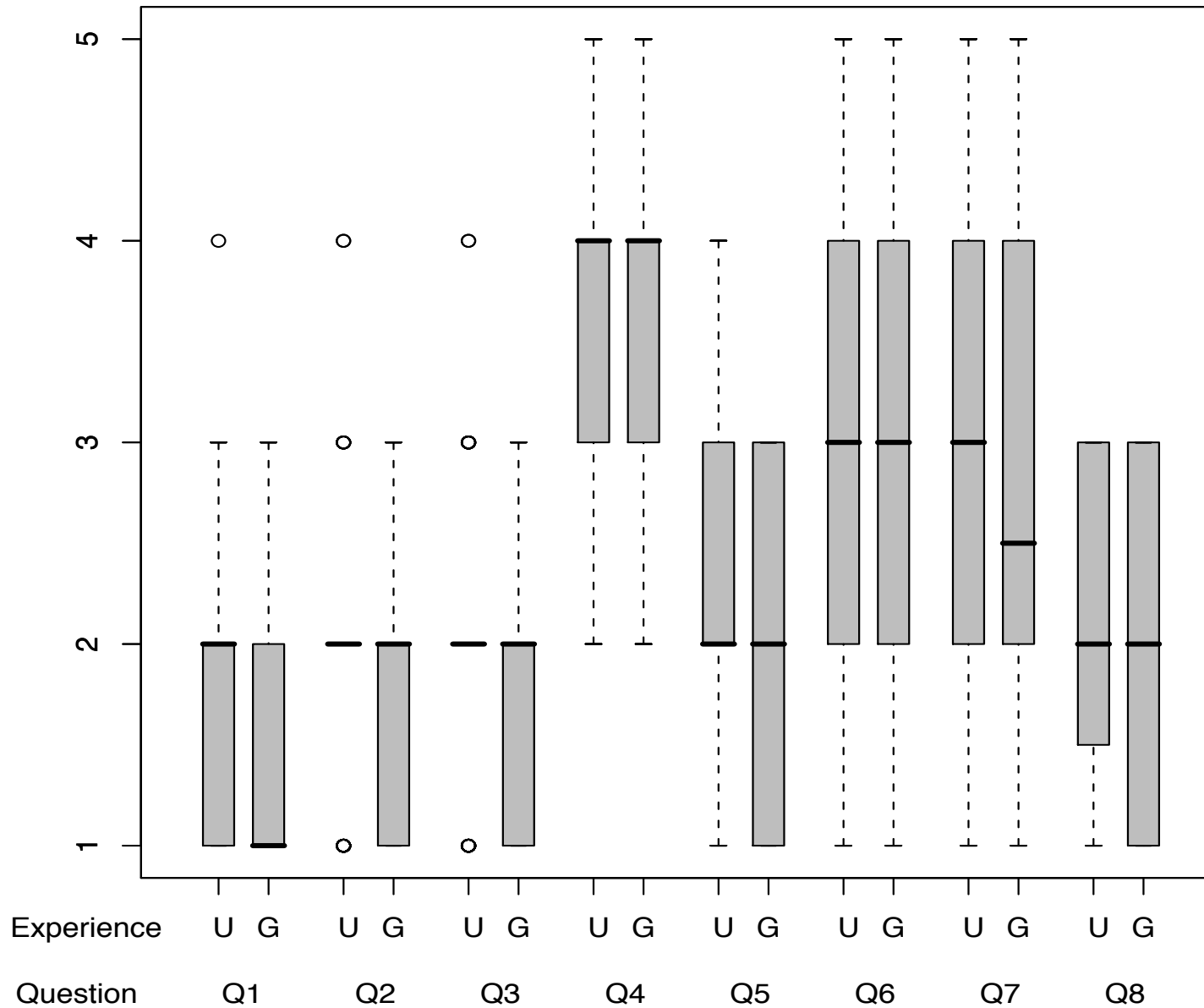
Id	Question
Q1	I had enough time to perform the lab task
Q2	The objectives of the lab were perfectly clear to me
Q3	The task I had to perform was perfectly clear to me
Q4	I experienced no major difficulties in performing the comprehension task
Q5	The domain of the system is clear to me
Q6	Evaluate your experience level on UML class diagrams
Q7	Evaluate your experience level on ER diagrams
Q8	Which was the easier notation to maintain?

Q1 – Q5: from 1 (strongly agree) to 5 (strongly disagree)

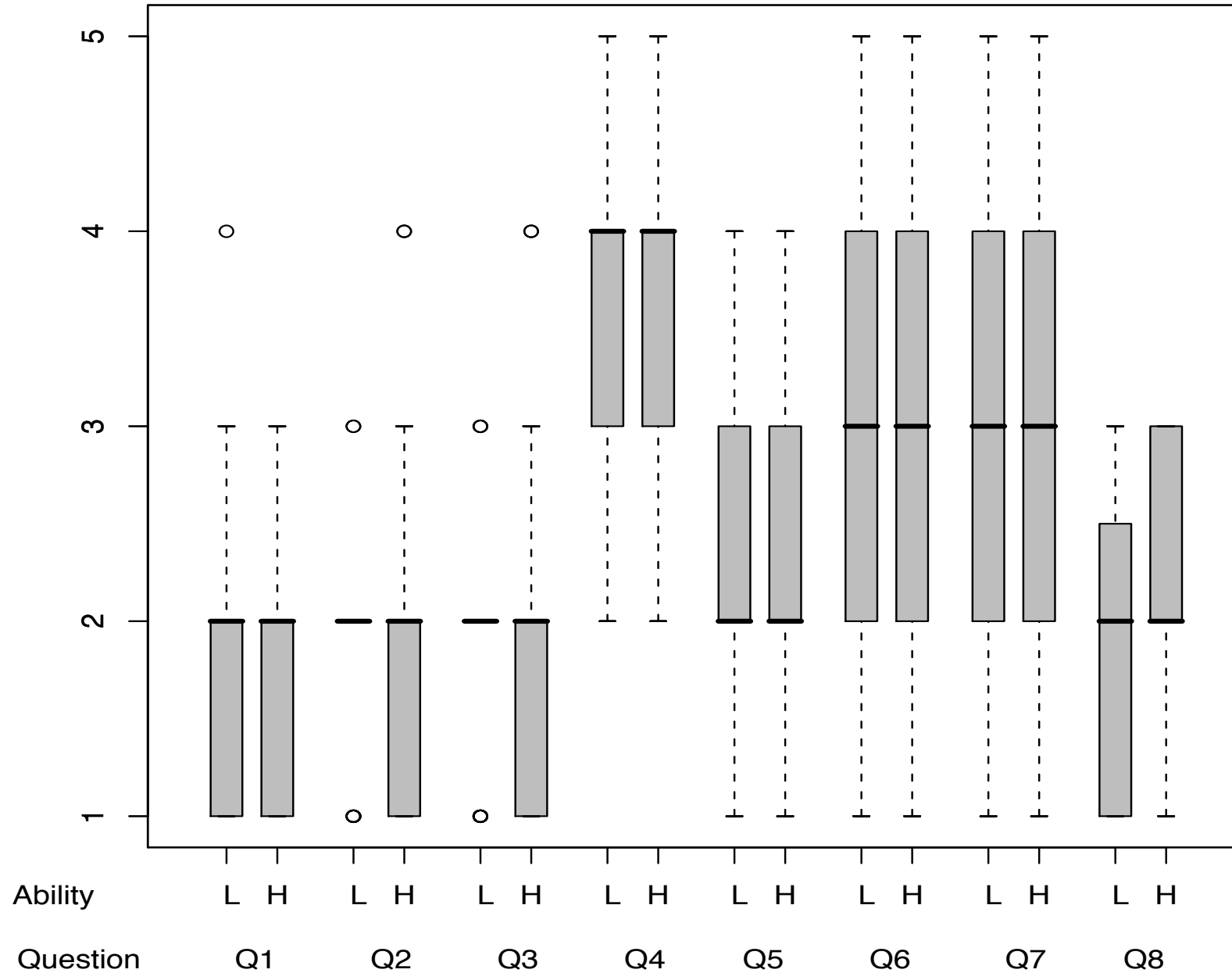
Q6 - Q7: from 1 (very high experience) to 5 (very low experience)

Q8: ER, UML CD

Final questionnaire: answers



Final questionnaire: answers



Threats to Validity (construct)

- Interactions between different treatments were mitigated by a proper design that allowed to separate the analysis of the different factors and of their interactions
- F-measure (aggregation of Precision and recall) well reflects the comprehension level
- To avoid social threats due to evaluation apprehension, students were not evaluated on their results achieved in the experiments.
- Subjects were not aware of the experimental hypotheses.

Threats to Validity (internal)

- learning effect experienced by subjects between labs mitigated thanks to the experiment design: subjects worked, over the two labs, on different tasks and using two different design methods
 - no significant effect of the **Lab**
- there was no abandonment, and as shown in the survey everything was clear in both the experiments.

Threats to Validity (external)

- students had participated to software projects, where they experienced software development and documentation, including database design
 - we plan to replicate the experiment with industrial subjects to corroborate the achieved findings.
- the complexity and size of data models to evolve are comparable to those of medium industrial projects

Threats to Validity (conclusion)

- proper tests were performed to statistically reject null hypotheses
- The F-measure allowed us to evaluate in an objective way the correctness level of subjects.
- survey questionnaires, mainly intended to get qualitative insights, were designed using standard ways and scales

Details in:

EMSE 2010

Empir Software Eng (2010) 15:455–492
DOI 10.1007/s10664-009-9127-7

Assessing the Support of ER and UML Class Diagrams during Maintenance Activities on Data Models

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Department of Mathematics and Informatics, University of Salerno

CSMR 2008

Abstract

We present the results of two controlled experiments carried out to compare the support given by the ER and UML class diagrams during the maintenance of data models. The experiments involved Master and Bachelor students performing maintenance tasks on data models represented by ER and UML class diagrams. The results reveal that the two notations give in general the same support. In particular, the correctness level achieved by a subject performing the task on data model represented by an ER diagram are comparable with the correctness level achieved by the same subject performing the task on a different data model represented by an UML class diagram. Moreover, by discriminating the levels of ability (high vs. low) and experience (graduate vs. undergraduate) of subjects we also provide some consideration about the influence of such factors on the correctness level achieved by subjects. In particular, we observe that UML class diagrams better support subjects with high ability than ER diagrams, while no difference can be observed considering subjects with low ability. Regarding the experience factor the results reveal no difference in the correctness level achieved by graduate and undergraduate students.

Index terms: Empirical Studies, Data Models, UML class diagrams, ER diagrams, Maintenance.

1. Introduction

Many graphical notations have been proposed in the literature to represent data models, such as the Entity-Relationship (ER) notation (and its extensions) [13] and Object-Oriented (OO) notations (e.g., UML class diagrams [19]). Very often, the choice of which notation has to be used comes from personal preferences or conventions, rather than how easy human resources understand or learn

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ers, ER is the most used notation for database conceptual modeling and still remains the *de facto* standard [13]. The success of the OO approach for programming achieved in the last years has encouraged the use of this approach also for modelling databases [20].

The 16th IEEE International Conference on Program Comprehension

Data Model Comprehension An Empirical Comparison of ER and UML Class Diagrams*

Andrea De

ICPC 2008

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Abstract

We present the results of two controlled experiments to compare ER and UML class diagrams, in order to find out which of the models provides better support during the comprehension of data models. The experiment involved Master and Bachelor students performing comprehension tasks on data models represented by ER or UML class diagrams. The achieved results show that UML class diagrams significantly improve the comprehension level achieved by subjects. Moreover, having different subjects with different levels of ability and experience allowed us to also make some considerations on the influence of such factors on the comprehension performances.

1. Introduction

A data model is a set of concepts that can be used to describe the structure of (i.e., data types, relationships, and constraints) and operations on a database [19]. Many graphical notations have been proposed in the literature to represent data models, such as the Entity-Relationship (ER) notation (and its extensions) [19] and Object-Oriented (OO) notations (e.g., UML class diagrams [26]). Very often, the choice of which notation has to be used comes from personal preferences or conventions, rather than how easy human resources understand or learn the notation.

Since many years, ER is the most used notation for database conceptual modelling and still remains the *de facto* standard [19]. The basic constructs are entities, attributes, and relationships, thus the notation is only focused on structural aspects and it does not take into account dynamic and behavioural aspects [19]. The success of the OO approach for programming achieved in the last years has encouraged

*The work described in this paper is supported by the project METAMORPHOS, funded by Ministero dell'Università e della Ricerca under grant PRIN-2006-2006098097.

the use of this approach also for modelling database [27]. However, widely used textbooks dedicate only few sections of a chapter to present the use of UML class diagrams as alternative to the ER diagrams [19].

As a consequence, in the majority of database courses the ER notation is used to design databases, while UML is presented only in software engineering courses. Thus, when students attend a course of software engineering they have a deeply knowledge of how to map ER diagrams onto relational schemas and do not find natural (and immediate) to apply similar mapping rules starting from UML class diagrams. Such a situation has a critical side effect. Indeed, in the context of software engineering courses during the development of software projects we observed that even if students use the UML notation for the analysis and design activities, they usually employ the ER notation to model the database of the software system. In particular, very often they translate the UML class diagram into an ER diagram maintaining the two representations of the data model. A similar situation has also been noticed in the industry, as confirmed by the results of a survey we have conducted with our industrial partners to understand which is the notation used in industrial contexts for database modelling. In particular, we found that that 4 of 13 contacted software companies usually employ both notations to represent the same database (see [11] for more details).

Although several empirical studies [5], [21], [27], [28] have been carried out to compare the ER and OO notations during database design, actual benefits given by the ER diagrams have not been highlighted to justify the real need of preferring this notation for data modelling. In this paper, we present the empirical results of two controlled experiments aiming at comparing ER and UML class diagrams during comprehension tasks on data models. The experiment was executed twice at the University of Salerno involving Bachelor and Master students, respectively. In the context of the experiments, subjects participated to two laboratory sessions where they performed comprehension tasks on data models represented by ER or UML class diagrams. Our

An experimental comparison of ER and UML class diagrams for data modelling

Andrea De Lucia · Carmine Gravino · Rocco Oliveto · Genoveffa Tortora

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© Springer Science+Business Media, LLC 2009
Editor: Erik Arisholm

Abstract We present the results of three sets of controlled experiments aimed at analysing whether UML class diagrams are more comprehensible than ER diagrams during data models maintenance. In particular, we considered the support given by the two notations in the comprehension and interpretation of data models, comprehension of the change to perform to meet a change request, and detection of defects contained in a data model. The experiments involved university students with different levels of ability and experience. The results demonstrate that using UML class diagrams subjects achieved better comprehension levels. With regard to the support given by the two notations during maintenance activities the results demonstrate that the two notations give the same support, while in general UML class diagrams provide a better support with respect to ER diagrams during verification activities.

Keywords Controlled experiments · Entity-relation diagrams · UML class diagrams · Design notations · Comprehension · Maintenance · Verification

The work described in this paper is supported by the project METAMORPHOS (MEthods and Tools for migrAting software systeMs towards web and service Oriented aRchitectures: exPerimental evaluation, usability, and techNOlogy tranSfer), funded by MiUR (Ministero dell'Università e della Ricerca) under grant PRIN-2006-2006098097.

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The Three Chapters: Chapter 2

After Chapter 1:

The support given by UML class diagrams in comprehension tasks is at least equal (or even higher) than the support given by ER diagrams

Still missing:

Qualitative and quantitative analyses concerning the identification of the graphical elements of one notation that are more comprehensible than the corresponding element in the other notation

These analyses could be used to:

- i. justify the need of preferring ER or UML CDs for data modeling
- ii. identify weaknesses of UML class diagrams

Design of experiments/replications

Goal Definition and Context

Goal

To analyse whether UML CDs are more comprehensible than ER diagrams during the comprehension and modification of data models

Fine-grained analysis comparing:

Entity, Primary Key/ID, Composite Attribute, Multi-value Attribute, Recursive relationship, 1:N Relationship, Ternary relationship, Generalization Relationship, Weak entity, M:N relationship

Students from the University of Salerno with different background:

- *bachelor students, 2nd year B.Sc. (n=37 in experiment -Comprehension)*
- *master students, 1st year M.Sc. (n=52 in 1st replication -Comprehension)*
- *fresher students, 1st year B.Sc. (n=67 in 2nd replication -Comprehension)*
- *master students, 1st year M.Sc. (n=28 in experiment -Modification)*

Design of experiments/replications (Cont.)

- Variable selection
- Hypotheses
- Experiment design

As in Chapter 1

Design of experiments/replications (Cont.)

Tasks

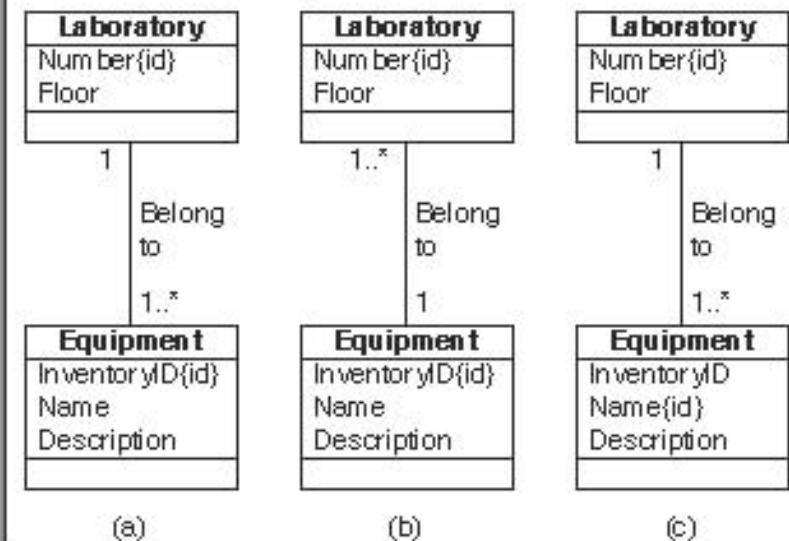
- **Questionnaire (Comprehension):**
 - 10 multiple choice questions
 - One for each building block
 - One or more correct answers
- **Questionnaire (Modification):**
 - 10 multiple choice questions
 - One for each building block
 - One correct answer

Q4 Let us focus on the classes Project and Company.

Which of the following statements is true:

- ☐ A company has a unique office
- ☐ A project has a unique office
- ☐ A company may have multiple offices

Q6 We want to specify the information about the equipments present in each laboratory. A laboratory can contain several equipments and each equipment is identified by an inventory code. A description and a name have also to be specified for an equipment. Indicate the part of the class diagram interested by this modification activity:



Design of experiments/replications (Cont.)

Tasks

- **Questionnaire (preferences):**
 - **A subjective questionnaire to specify preferences between notations, for each building block**
 - [] Entity-Relationship (ER)
 - [] Class Diagram (CD)
 - [] No preference

Final questionnaire

- S_1 : I had enough time to perform the tasks
- S_2 : The task objectives were perfectly clear to me
- S_3 : The tasks I performed was perfectly clear to me
- S_4 : Judging the difficulty of the assigned task

where S_1 , S_2 , and S_3 expected closed answers according to the Likert scale [?] going from 1 (strongly disagree) to 5 (strongly agree), while for S_4 the answer ranges from 1 (very low) to 5 (very high).

Design of experiments/replications (Cont.)

Procedure and Data Analysis

- **Training:** on ER and CDs and on experiment execution
- **Execution:** Tasks performed individually
- **Analysis:**

Behavior of the dependent variable

Descriptive statistics

Differences between treatments

Paired Wilcoxon test, each subject performed a task on two different models with the two possible treatments

Preferences between treatments

Histograms

Results: Comprehension (Cont.)

**Comprehension level
(descriptive statistics)**

Subjects	ER			CD		
	Mean	Median	St. Dev.	Mean	Median	St. Dev.
Fresher	0.801	1.000	0.307	0.816	1.000	0.280
Bachelor	0.849	1.000	0.242	0.845	1.000	0.278
Master	0.849	1.000	0.277	0.838	1.000	0.272

Both notations **provide comparable support** when performing comprehension activities on data models

Wilcoxon test

NO significant differences

Subjects	CD\$FM - ER\$FM			p-value	effect size
	Mean	Median	St. Dev.		
Fresher	0.014	0.000	0.404	0.343	0.037
Bachelor	0.003	0.000	0.330	0.420	-0.011
Master	-0.012	0.000	0.383	0.817	-0.030

Results: Comprehension (Cont.)

Fine-grained level analysis

Overall *performance equilibrium* confirmed

Method Element		Fresher			Bachelor			Master		
		Mean	Median	St. Dev.	Mean	Median	St. Dev.	Mean	Median	St. Dev.
ER	Entity	0.887	1.000	0.260	0.936	1.000	0.125	0.872	1.000	0.281
	Primary Key/ID	0.784	1.000	0.406	0.955	1.000	0.179	0.907	1.000	0.277
	Composite attribute	0.883	1.000	0.159	0.897	1.000	0.146	0.920	1.000	0.140
	Multi-value attribute	0.859	1.000	0.195	0.847	1.000	0.168	0.862	1.000	0.213
	Recursive relationship	0.779	1.000	0.301	0.757	0.667	0.224	0.817	1.000	0.243
	Relationship cardinality	0.875	1.000	0.240	0.892	1.000	0.158	0.929	1.000	0.179
	Ternary relationship	0.741	1.000	0.347	0.828	1.000	0.220	0.804	1.000	0.321
	Generalization IS-A	0.684	0.667	0.369	0.734	1.000	0.363	0.712	1.000	0.379
	Weak entity	0.725	0.800	0.266	0.767	1.000	0.305	0.747	0.900	0.329
	M:N relationship	0.789	1.000	0.368	0.865	1.000	0.319	0.923	1.000	0.244
CD	Entity	0.961	1.000	0.108	0.937	1.000	0.234	0.926	1.000	0.145
	Primary Key/ID	0.875	1.000	0.296	0.937	1.000	0.234	0.926	1.000	0.246
	Composite attribute	0.742	0.667	0.255	0.781	0.800	0.251	0.815	1.000	0.308
	Multi-value attribute	0.775	0.667	0.259	0.788	0.667	0.257	0.801	0.667	0.209
	Recursive relationship	0.767	1.000	0.323	0.856	1.000	0.226	0.806	0.800	0.210
	Relationship cardinality	0.865	1.000	0.261	0.856	1.000	0.320	0.906	1.000	0.150
	Ternary relationship	0.827	1.000	0.265	0.888	1.000	0.150	0.855	1.000	0.162
	Generalization IS-A	0.828	1.000	0.225	0.838	1.000	0.290	0.804	1.000	0.328
	Weak entity	0.629	0.667	0.407	0.611	0.667	0.407	0.608	0.733	0.447
	M:N relationship	0.890	1.000	0.162	0.955	1.000	0.179	0.929	1.000	0.212

Results: Comprehension (Cont.)

Wilcoxon test (where ER performs better than CDs)

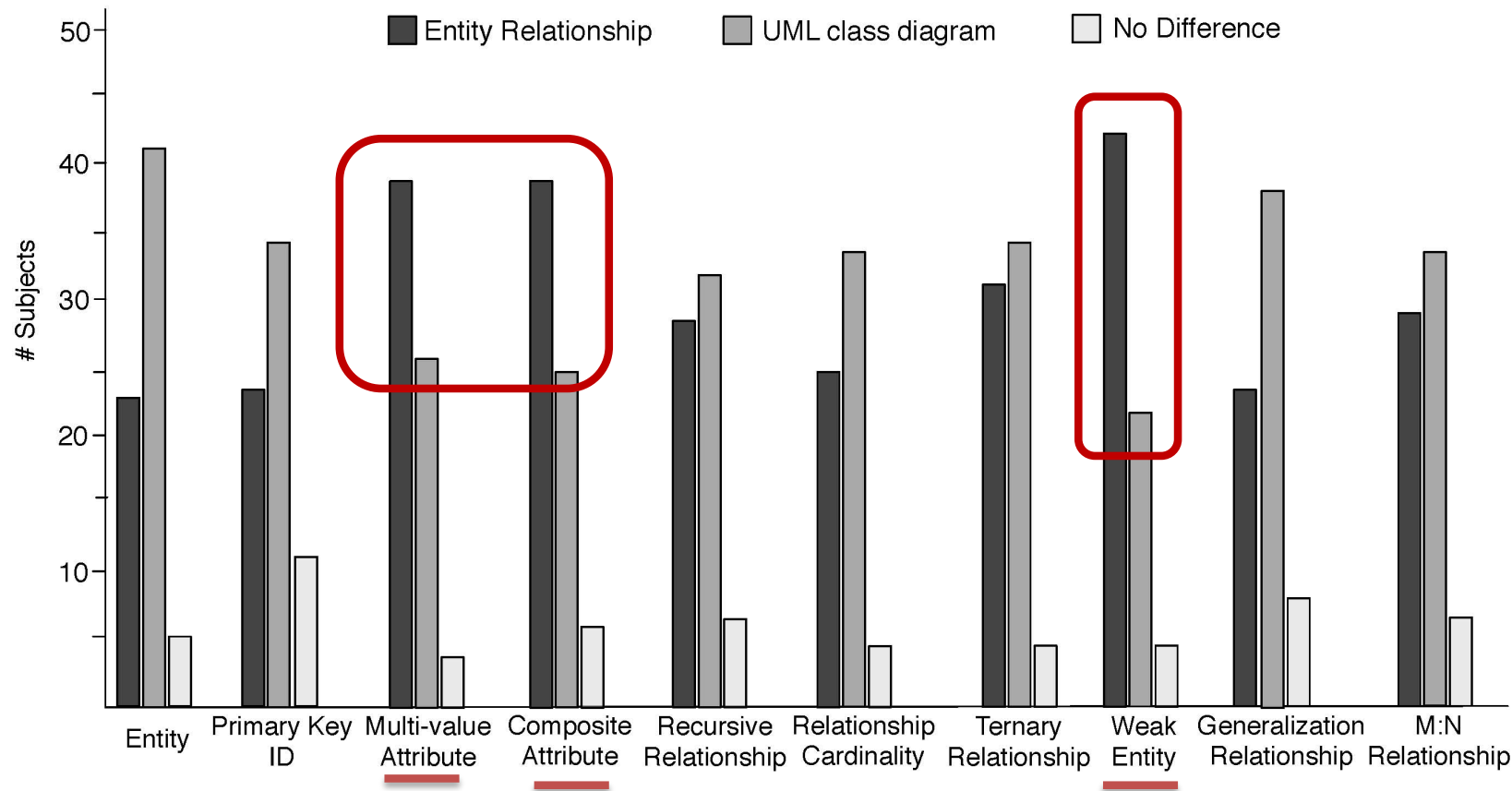
Element	Fresher					Bachelor					Master				
	ER\$FM - CD\$FM			p-value	effect size	ER\$FM - CD\$FM			p-value	effect size	ER\$FM - CD\$FM			p-value	effect size
	Mean	Median	St. Dev.			Mean	Median	St. Dev.			Mean	Median	St. Dev.		
Entity	-0.059	0.000	0.262	0.983	-0.257	-0.036	0.000	0.153	0.599	-0.032	-0.054	0.000	0.309	0.796	-0.161
Primary Key/ID	-0.091	0.000	0.517	0.927	-0.166	-0.027	0.000	0.198	0.415	0.059	-0.019	0.000	0.388	0.660	-0.049
Composite attribute	0.141	0.000	0.303	0.000	0.490	0.116	0.000	0.306	0.022	0.380	0.105	0.000	0.304	0.012	0.343
Multi-value attribute	0.085	0.000	0.316	0.014	0.269	0.059	0.000	0.324	0.141	0.180	0.061	0.000	0.311	0.080	0.196
Recursive relationship	0.012	0.000	0.401	0.433	0.024	-0.010	0.000	0.287	0.983	-0.345	0.011	0.000	0.308	0.536	0.037
Relationship cardinality	0.009	0.000	0.358	0.439	0.028	-0.009	0.000	0.200	0.446	0.094	0.023	0.000	0.224	0.258	0.103
Ternary relationship	-0.086	0.000	0.471	0.897	-0.184	-0.042	0.000	0.266	0.869	-0.221	-0.050	0.000	0.368	0.720	-0.135
Generalization IS-A	-0.145	0.000	0.421	0.999	-0.388	-0.104	0.000	0.476	0.995	-0.217	-0.093	0.000	0.526	0.992	-0.177
Weak entity	0.096	0.000	0.457	0.027	0.211	0.156	0.000	0.504	0.045	0.309	0.139	0.000	0.590	0.049	0.234
M:N relationship	-0.105	0.000	0.379	0.972	-0.249	-0.045	0.000	0.334	0.942	-0.252	-0.006	0.000	0.313	0.562	-0.020

bold if ER comprehension level statistically higher than CD comprehension level

CDs weaknesses are identified in **Composite attributes**, **Multi-value attributes** and **Weak entities**

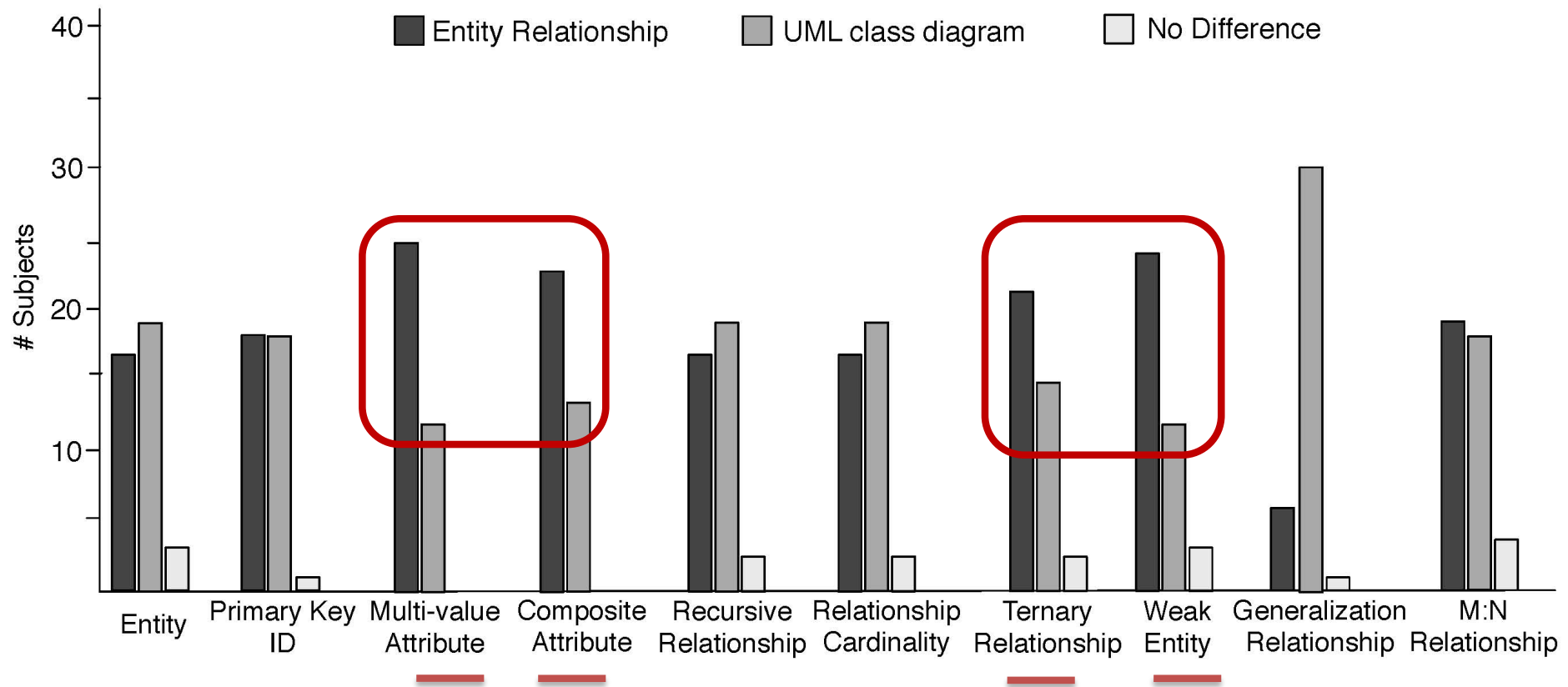
Results: Comprehension (Cont.)

- Preferences expressed by Fresher students



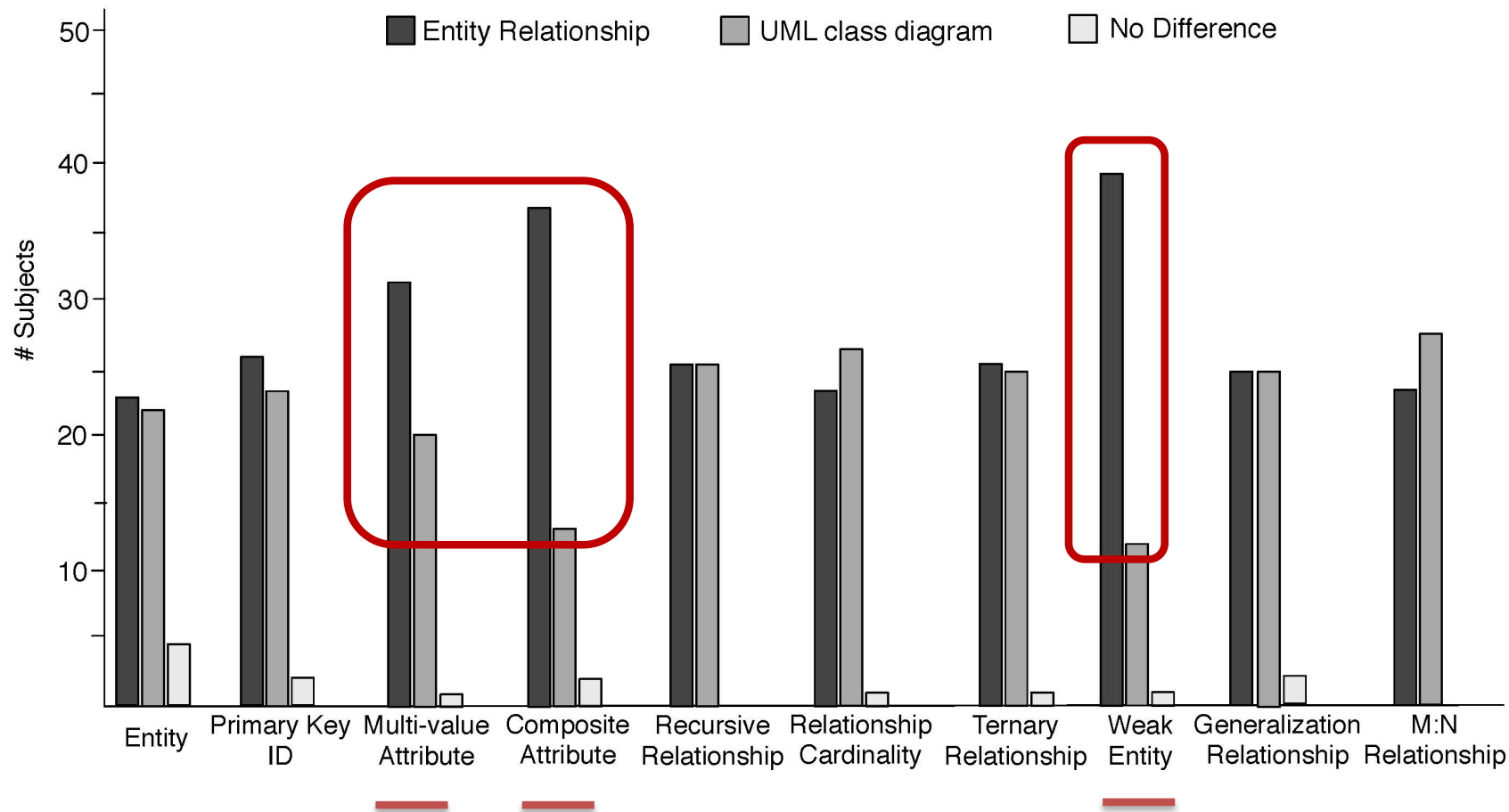
Results: Comprehension (Cont.)

- Preferences expressed by Bachelor students



Results: Comprehension (Cont.)

- Preferences expressed by Master students



Results: Modification (Cont.)

Wilcoxon test (where ER performs better than CDs)

Fine-grained level analysis (descriptive statistics)

Method	Element	Mean	Median	St. Dev.
ER	Entity	0.750	1.000	0.441
	Primary Key/ID	0.821	1.000	0.390
	Composite attribute	0.714	1.000	0.460
	Multi-value attribute	0.679	1.000	0.476
	Recursive relationship	0.714	1.000	0.460
	Relationship cardinality	0.857	1.000	0.356
	Ternary relationship	0.607	1.000	0.497
	Generalization IS-A	0.643	1.000	0.488
	Weak entity	0.821	1.000	0.390
	M:N relationship	0.714	1.000	0.460
	Entity	0.929	1.000	0.262
	Primary Key/ID	0.893	1.000	0.315
CD	Composite attribute	0.678	1.000	0.476
	Multi-value attribute	0.464	1.000	0.508
	Recursive relationship	0.571	1.000	0.504
	Relationship cardinality	0.929	1.000	0.262
	Ternary relationship	0.893	1.000	0.315
	Generalization IS-A	0.786	1.000	0.418
	Weak entity	0.536	1.000	0.508
	M:N relationship	0.821	1.000	0.390

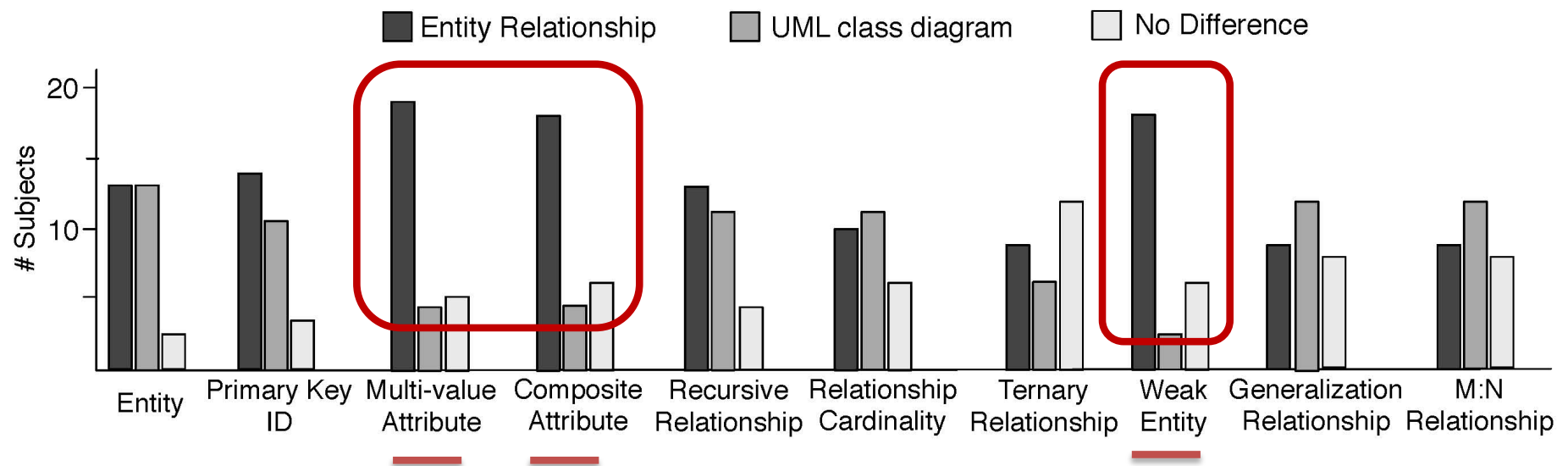
Element	ER vs. CD			p-value	effect size
	Mean	Median	St. Dev.		
Entity	0.179	0.000	0.476	0.976	0.375
Primary Key/ID	0.071	0.000	0.539	0.784	0.132
Composite attribute	0.026	0.000	0.602	0.406	-0.052
Multi-value attribute	-0.214	0.000	0.630	0.045	-0.340
Recursive relationship	-0.143	0.000	0.705	0.151	-0.203
Relationship cardinality	0.071	0.000	0.378	0.885	0.189
Ternary relationship	0.286	0.000	0.600	0.991	0.476
Generalization IS-A	0.143	0.000	0.651	0.885	0.220
Weak entity	-0.286	0.000	0.600	0.012	-0.476
M:N relationship	0.107	0.000	0.629	0.830	0.170

The value is in **bold** when the ER comprehension level is statistically higher than CD comprehension level

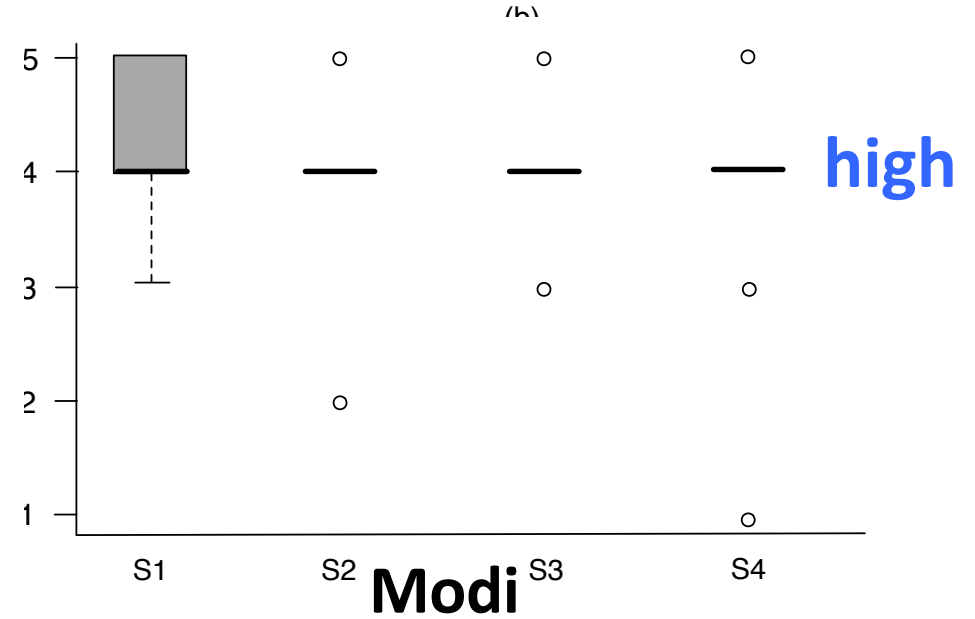
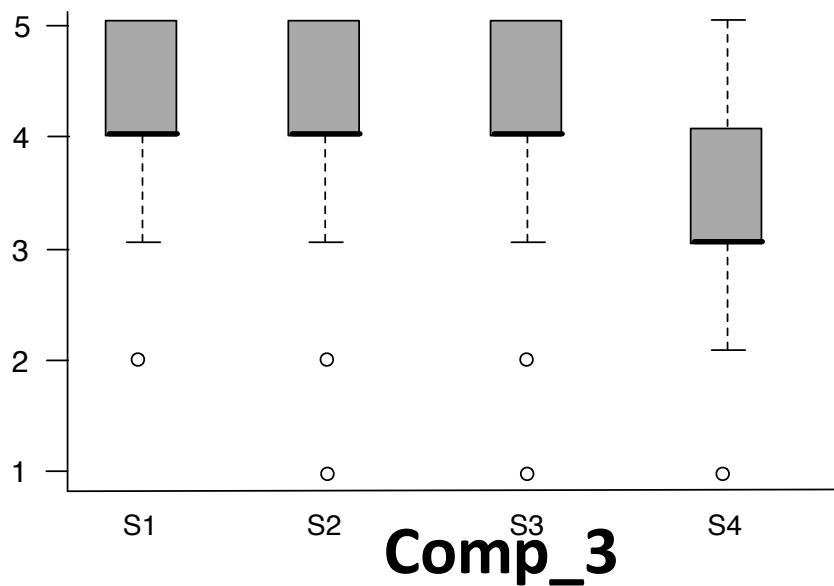
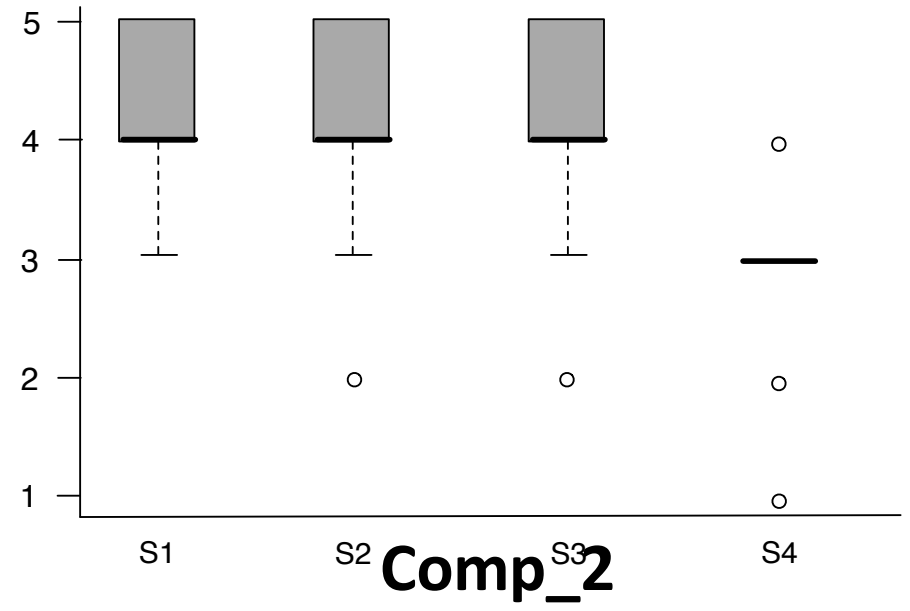
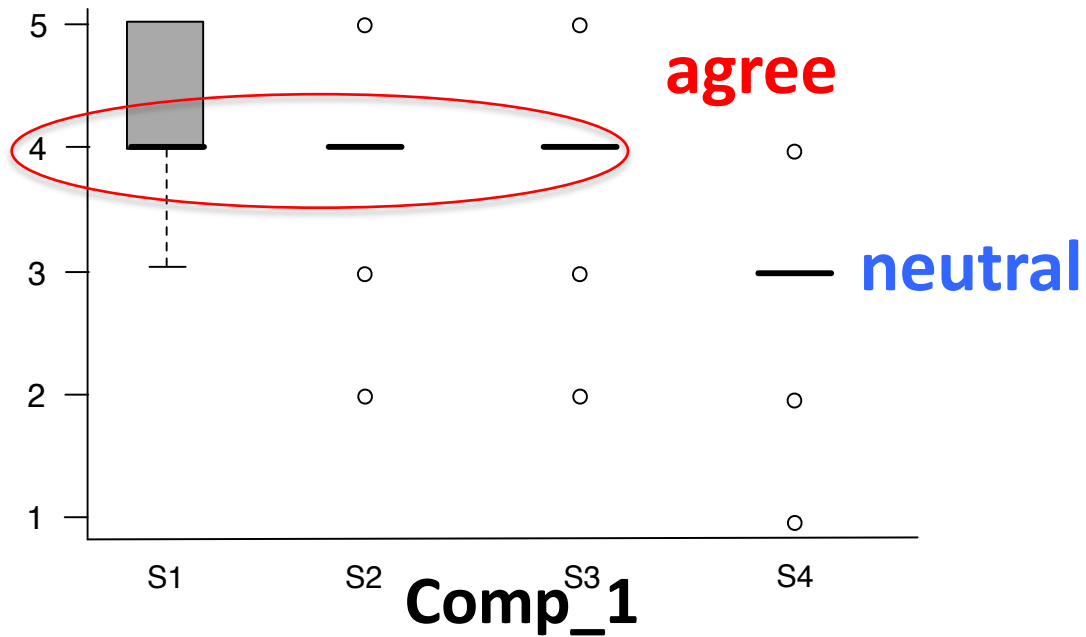
CDs weaknesses are identified in **Multi-value attributes** and **Weak entities**

Results: Modification (Cont.)

- Preferences expressed by Master students



Final questionnaire answers



Details in:

SOSYM

Software and Systems Modeling manuscript No.
(will be inserted by the editor)

Identifying the Weaknesses of UML Class Diagrams during Data Model Comprehension

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Abstract. In this paper we present an experiment and two replications aimed at comparing the support provided by ER and UML class diagrams during comprehension activities by focusing on the single building blocks of the two notations. This kind of analysis can be used to identify weakness in a notation and/or justify the need of preferring ER or UML for data modeling. The results reveal that UML class diagrams are generally more comprehensible than ER diagrams, even if the former has some weaknesses related to three building blocks, i.e., multi-value attribute, composite attribute, and weak entity. These findings suggest that a UML class diagram extension should be considered to overcome these weaknesses and improve the comprehensibility of the notation.

1 Introduction

A data model is a set of concepts that can be used to describe both the structure of and the operations on a database [1]. It represents the output of data modeling (or conceptual design), an activity that aims at creating a conceptual schema in a diagrammatic form and facilitating the communication between developers and users [1]. Understanding and interpreting data models represents a fundamental activity from the earliest stages of software development, e.g., requirement analysis. Thus, a comprehensive notation is really desirable to avoid misunderstanding that can lead to the introduction of errors very expensive to remove in the later phases of the software development. A comprehensive notation is also desirable during software maintenance, since it facilitates the comprehension activities that have to be performed to understand the design of the system before the analysis and the implementation of a change request.

Entity-Relationship (ER) and its extensions are the most used notations for database conceptual modeling and still remains the *de facto* standard [1]. The success of the Object-Oriented (OO) approach for software development has encouraged the use of this approach also for database modeling [2]. In particular, UML class diagrams can be used to represent the conceptual schema of the whole

A Fine-Grained Analysis of the Support Provided by UML Class Diagrams and ER Diagrams During Data Model Maintenance*

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Received: date / Revised version: date

Abstract This paper presents the results of an empirical study aiming at comparing the support provided by ER and UML class diagrams during maintenance of data models. We performed one controlled experiment and two replications that focused on comprehension activities (the first activity in the maintenance process) and another controlled experiment on modification activities related to the implementation of given change requests. The results achieved were analyzed at a fine-grained level aiming at comparing the support given by each single building block of the two notations. Such an analysis is used to identify weaknesses (i.e., building blocks not easy to comprehend) in a notation and/or can justify the need of preferring ER or UML for data modeling. The analysis revealed that UML class diagrams generally provided a better support for both comprehension and modification activities performed on data models as compared to ER diagrams. Nevertheless, the former has some weaknesses related to three building blocks, i.e., Multi-value attribute, Composite attribute, and Weak entity. These findings suggest that an extension of UML class diagrams should be considered to overcome these weaknesses and improve the support provided by UML class diagrams during maintenance of data models.

1 Introduction

Data modeling is a process used to define and analyze data requirements needed to support the business processes within the scope of corresponding information systems in organizations [21]. This process usually involves

data modelers that work closely with business stakeholders and potential users of the information system.

The output of the data modeling process is represented by a *data model*. The data model describes both the structure of and the operations on a database in a diagrammatic form aiming at facilitating the communication between the stakeholders involved in the data modeling process [21].

Understanding and interpreting data models represents a fundamental activity from the earliest stages of software development, e.g., requirement analysis. Thus, a comprehensive notation is really desirable to avoid misunderstanding that can lead to the introduction of errors very expensive to remove in the later phases of the software development.

A comprehensive notation is also desirable to facilitate the maintenance of the produced data model. Indeed, data models should be considered living documents that will change in response to a changing business. Thus, a comprehensive notation can facilitate the comprehension activities that have to be performed to understand the data model before the analysis and the implementation of a change request.

Entity-Relationship (ER) and its extensions are the most used notations for database conceptual modeling and still remains the *de facto* standard [21]. The success of the Object-Oriented (OO) approach for software development has encouraged the use of this approach also for database modeling [26]. Specifically, UML class diagrams can be used to represent the conceptual schema of the whole software system, so the same notation can be used to model the functionality of the system as well as to represent its data. The structural constructs of the UML class diagram which represents the data structure is somewhat equivalent to Extended ER (EER) representation (e.g., object classes considered equivalent to entity and relationship types). The functionality is represented through “methods” that are attached to the object classes.

* This paper is an extension of the work “Identifying the Weaknesses of UML Class Diagrams during Data Model Comprehension” appeared in the *Proceedings of the 14th International Conference on Model Driven Engineering Languages and Systems*, Wellington, New Zealand, pages 168–182, 2011. LNCS Press.

Chapter 3

After Chapters 1 and 2:

- The results of all the studies do not allow to identify a notation that is better than the other in all the circumstances
- UML class diagrams have been shown to be generally more comprehensible and easier to modify than ER diagrams, but also characterized by three weaknesses related to the representation of **Composite attribute**, **Multi-value attribute**, and **Weak entity**.

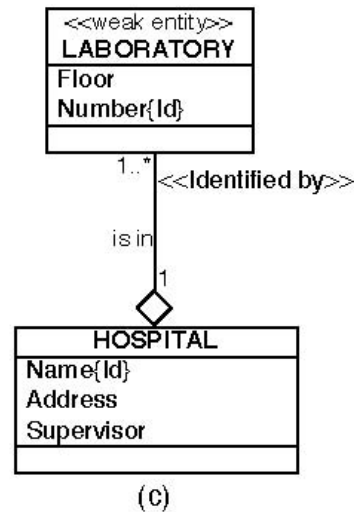
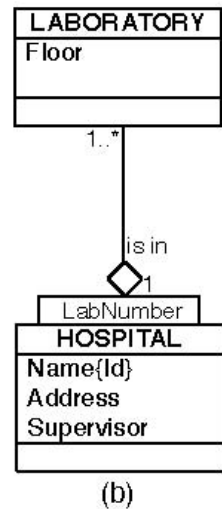
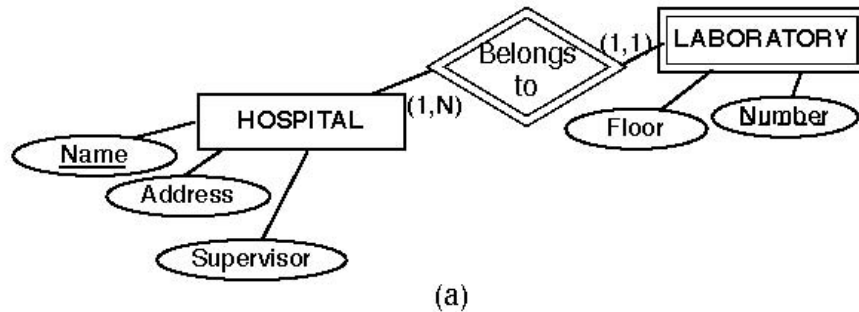
Still missing:

- to overcome the weakness of UML class diagrams
 - we have proposed and investigated the use of stereotypes (extension mechanism of UML)

These analyses could be used:

to make UML class diagrams more specific for data modeling

The proposed extension



Note:

As for Chapter 2
we exploit CD for data
modeling as made by
Elmasri and Navathe in:
*Fundamentals of Database
Systems*.
Addison Wesley, 2003

Fig. 1: The Weak Entity specified using ER notation (a), classical UML class diagram notation (b), and the proposed stereotypes (c).

The proposed extension (cont.)

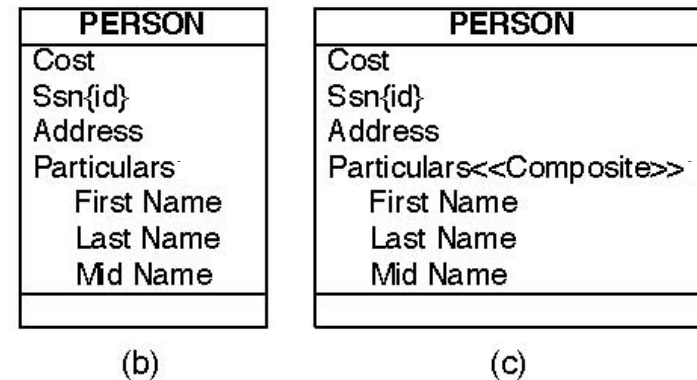
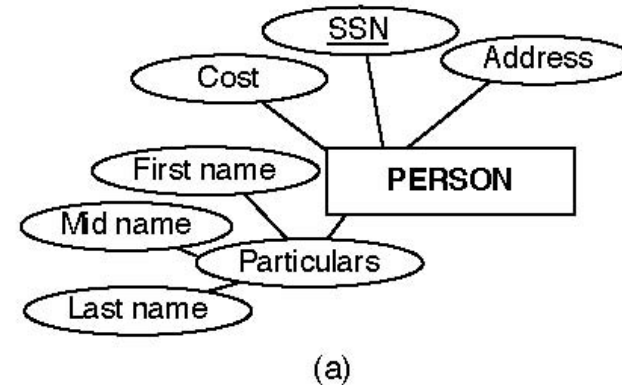
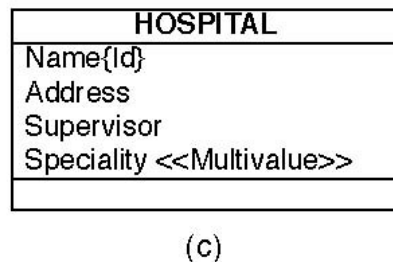
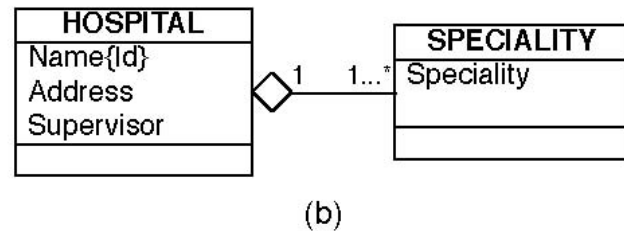
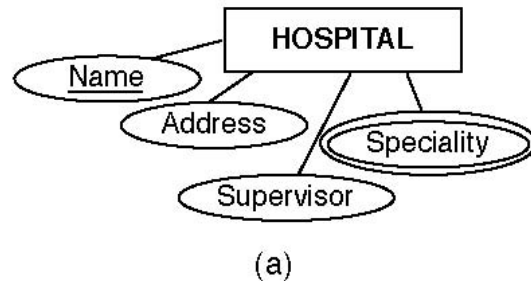


Fig. 2: The Multi-value attribute specified using ER notation (a), classical UML class diagram notation (b), and the proposed stereotypes (c).

Fig. 3: The Composite attribute specified using ER notation (a), classical UML class diagram notation (b), and the proposed stereotypes (c).

Design of experiments/replications

Goal Definition and Context

Goal

To analyse whether UML CDs are more comprehensible than ER diagrams during the comprehension and modification of data models

Fine-grained analysis comparing:

Entity, Primary Key/ID, Composite Attribute, Multi-value Attribute, Recursive relationship, 1:N Relationship, Ternary relationship, Generalization relationship, Weak entity, M:N relationship

Students from the University of Salerno (IT) and University of Castilla-La Mancha (SP), with different background:

- (IT) bachelor students, 2nd year B.Sc. (n=56 in experiment -Comprehension)
- (IT) master students, 1st year M.Sc. (n=21 in 1st replication -Comprehension)
- (SP) bachelor students, 2nd year B.Sc.(n=90 in 2nd replication -Comprehension)
- (IT) bachelor students, 2nd year B.Sc. (n=45 in experiment -Modification)
- (IT) master students, 1st year M.Sc. (n=27 in experiment -Modification)

Design of experiments/replications (cont.)

- Variable selection
- Systems:
 - EasyClinic
 - Company,
 - University

As in Chapter 2

Design of experiments/replications (cont.)

- **Research questions**

- RQ1: Does **CD_EX** provide a better support than **CD** in ***comprehending*** and ***modifying*** data models?
 - *if CD_EX provides worst support than CD we can mark the experimented extension as fully useless*
- RQ2: Does **CD_EX** provide a better support than **ER** in ***comprehending*** and ***modifying*** data models?
 - *to verify if the three weaknesses identified in Chapter 2 for the CD notation with respect to the ER one are overcome by the extended CD notation*

Design of experiments/replications (cont.)

Null Hypotheses:

- Hn0: there is no difference in the *comprehension* level of participants using the **CD** notation or the **CD_EX** notation
- Hn1: there is no difference in the *modification* level of participants using the **CD** notation or the **CD_EX** notation
- Hn2: there is no difference in the *comprehension* level of participants using the **ER** notation or the **CD_EX** notation
- Hn3: there is no difference in the *modification* level of participants using the **ER** notation or the **CD_EX** notation
- **Benefit derived by the CD_EX when:**
 - null hypotheses are rejected for the three building blocks identified as weaknesses and the significant difference is in favor of **CD_EX**, or
 - we are not able to reject our null hypothesis for the same three building blocks (i.e., ER does not outperform **CD_EX**).

Design of experiments/replications (cont.

- Procedure and Data Analysis
- Tasks

As in Chapter 2

Results: Comprehension

Descriptive statistics (where ER performs better than CDs)

- The three weaknesses of CD confirmed in all the experiments
 - difference of performances between ER and CD for the three weaknesses is on average 9% in favor of ER
- For the other seven building blocks there is a substantially equilibrium between ER and CD
 - for the remaining seven building blocks CD performs on average 0.3% better than ER.
- Note that this trend is confirmed with small variations among all the three experiments

Results: Comprehension (Cont.)

Wilcoxon test

TABLE 4: Comprehension activities: Wilcoxon paired test (CD_EX vs CD).

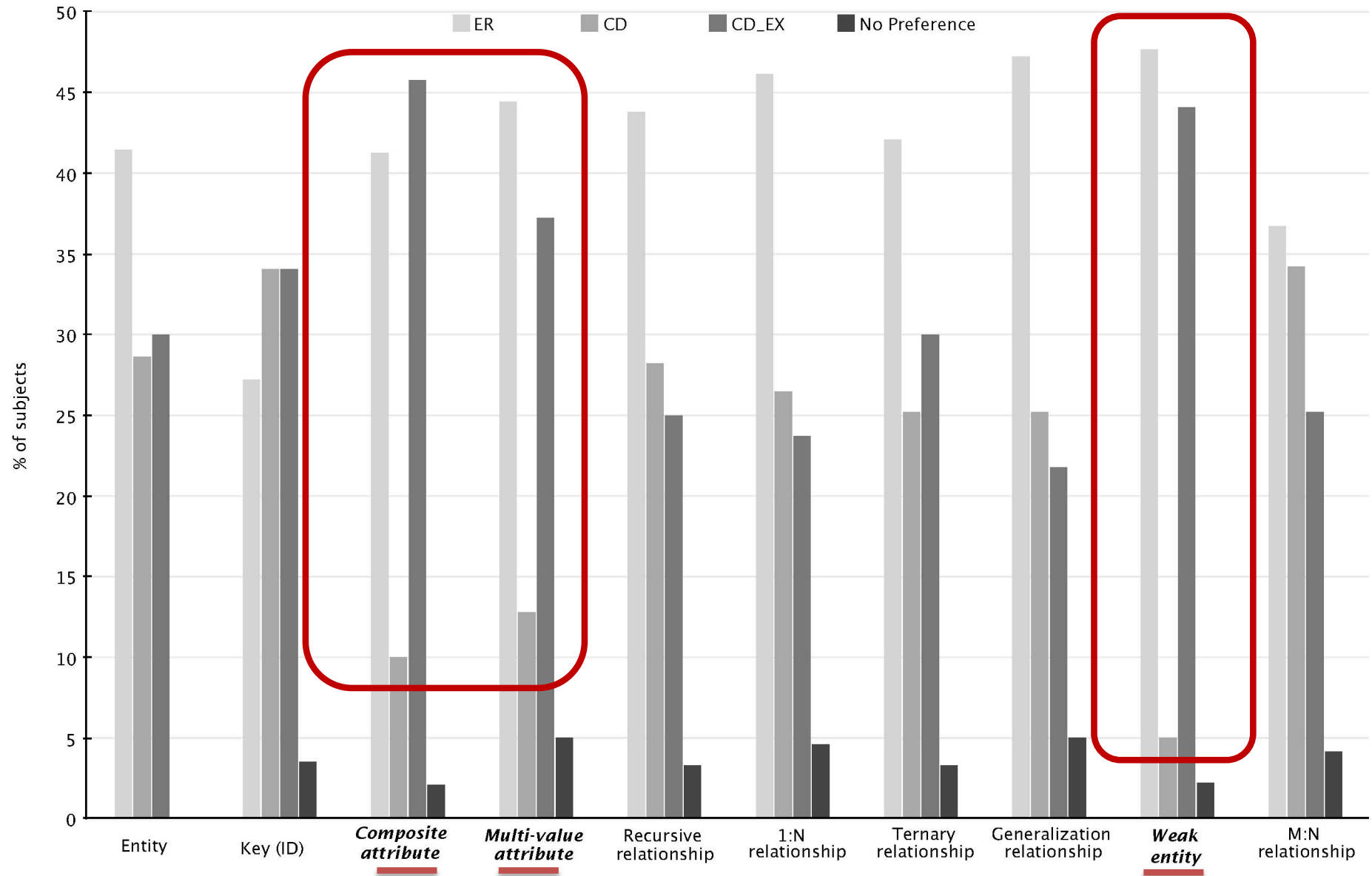
Element	Bachelor (Italy - 56 participants)	Master (Italy - 21 participants)	Bachelor (Spain - 90 participants)
Entity	0.9708	0.3478	0.07758
Primary key (ID)	1.0000	1.0000	0.1585
Composite attribute	0.0073 (CD_EX)	0.0192 (CD_EX)	0.7534
Multi-value attribute	0.0031 (CD_EX)	0.0041 (CD_EX)	0.0011 (CD_EX)
Recursive Relationship	0.4989	0.2166	0.6937
1:N Relationship	0.1702	0.1584	0.8297
Ternary relationship	0.9741	0.6577	0.6596
Generalization relationship	0.0261 (CD_EX)	0.8589	0.2996
Weak entity	0.0102 (CD_EX)	0.0121 (CD_EX)	0.0096 (CD_EX)
M:N Relationship	0.6865	1.000	0.7038

TABLE 5: Comprehension Activities: Wilcoxon paired test (CD_EX vs ER).

Element	Bachelor (Italy - 56 participants)	Master (Italy - 21 participants)	Bachelor (Spain - 90 participants)
Entity	0.3458	1.0000	0.2834
Primary key (ID)	0.5743	1.0000	0.1585
Composite attribute	0.1138	0.0305 (CD_EX)	0.6727
Multi-value attribute	0.2047	0.0480 (CD_EX)	0.9179
Recursive Relationship	0.4579	0.9073	0.3451
1:N Relationship	0.6549	0.1646	0.6778
Ternary relationship	0.3209	0.9294	0.2417
Generalization relationship	0.0603	0.1602	0.0570
Weak entity	<0.0001 (CD_EX)	<0.0001 (CD_EX)	0.4453
M:N Relationship	0.2171	0.4484	0.3080

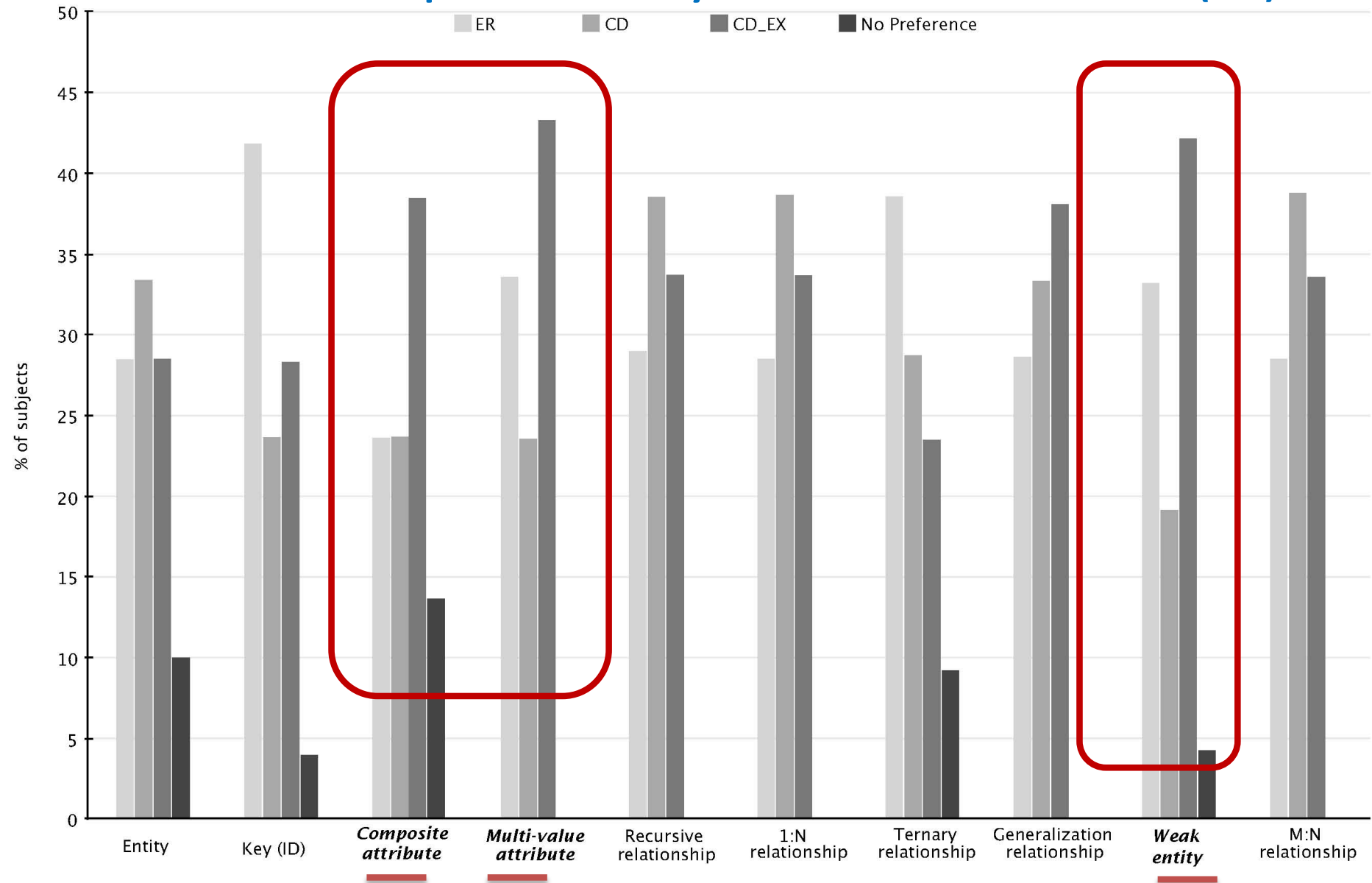
Results: Comprehension (Cont.)

- Preferences expressed by Bachelor students (IT)



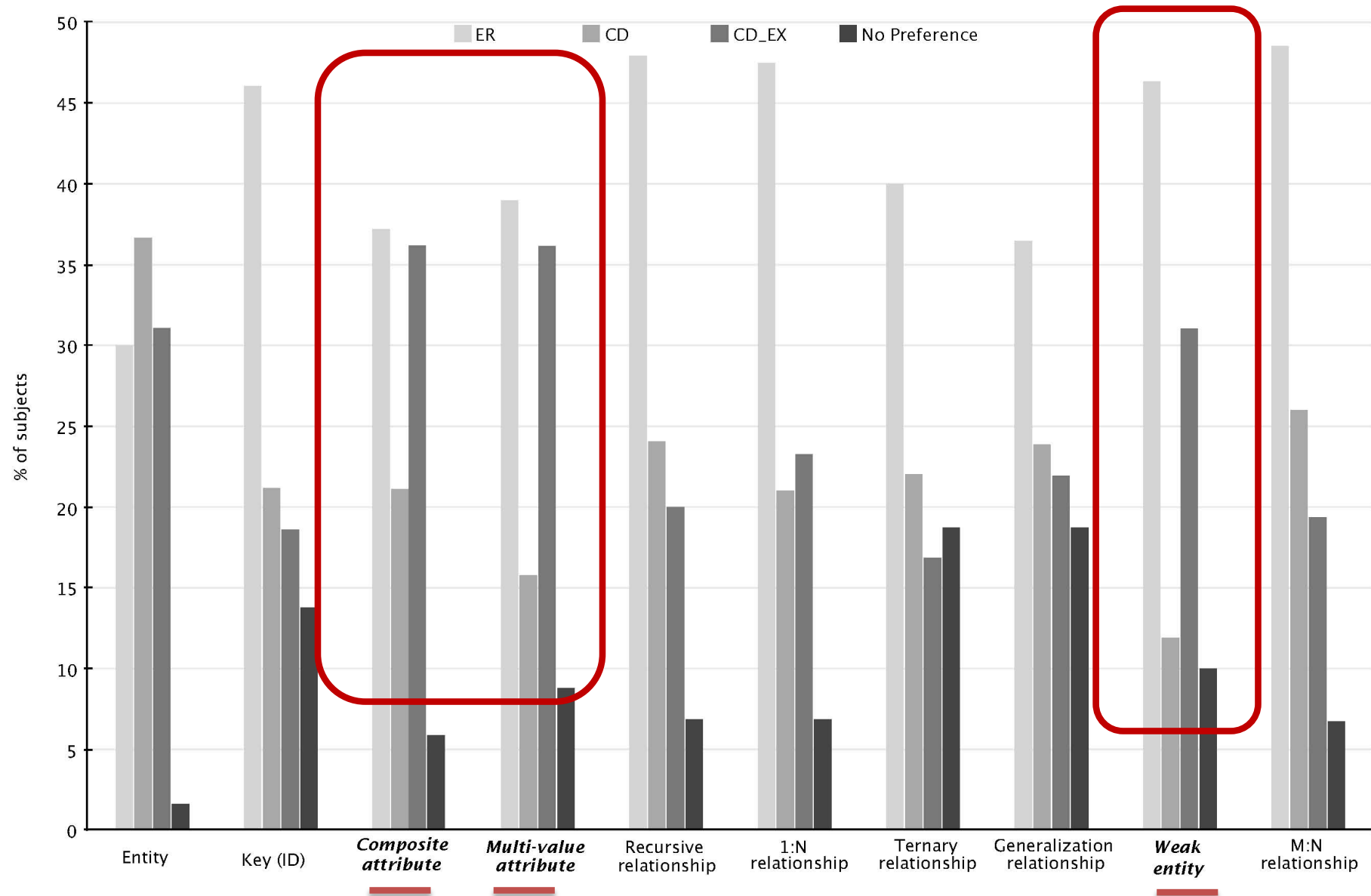
Results: Comprehension (Cont.)

- Preferences expressed by Master students (IT)



Results: Comprehension (Cont.)

- Preferences expressed by Bachelor students (SP)



Results: Modification

- On the three weaknesses identified in Chapter 2, ER achieved an average modification level higher than CD of about 17%.
- The situation is totally upside down for the remaining seven building blocks:
 - CD achieved a higher average modification level of about 10%.

Results: Modification (Cont.)

Wilcoxon test

TABLE 8: Modification activity: Wilcoxon paired test (CD_EX vs CD).

Element	Bachelor (Italy - 45 participants)	Master (Italy - 27 participants)
Entity	0.2200	0.7472
Primary key (ID)	0.1377	1.0000
<i>Composite attribute</i>	0.0210 (CD_EX)	0.0281 (CD_EX)
<i>Multi-value attribute</i>	0.0440 (CD_EX)	0.0281 (CD_EX)
Recursive Relationship	1.0000	0.6542
1:N Relationship	0.5420	1.0000
Ternary relationship	0.2981	0.5244
Generalization relationship	0.1318	0.2343
<i>Weak entity</i>	<0.0001 (CD_EX)	0.0198 (CD_EX)
M:N Relationship	0.6180	0.6991

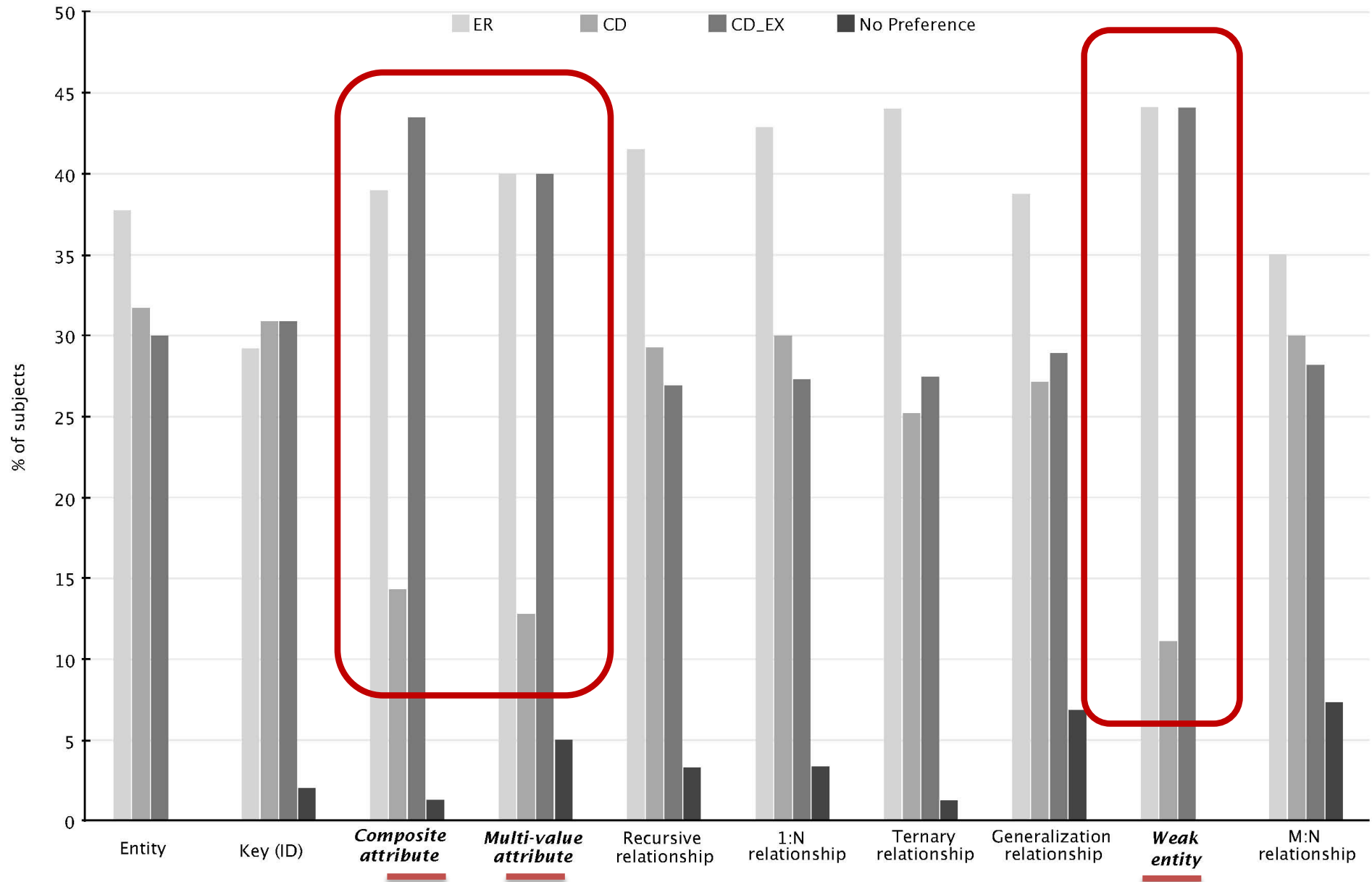
TABLE 9: Modification activity: Wilcoxon paired test (CD_EX vs ER).

Element	Bachelor (Italy - 45 participants)	Master (Italy - 27 participants)
Entity	0.7326	1.0000
Primary key (ID)	0.7934	0.6991
<i>Composite attribute</i>	0.3159	0.1690
<i>Multi-value attribute</i>	0.1105	0.4563
Recursive Relationship	0.8264	0.0281 (CD_EX)
1:N Relationship	0.7833	0.2838
Ternary relationship	<0.0001 (CD_EX)	0.3501
Generalization relationship	0.0846	0.0385 (CD_EX)
<i>Weak entity</i>	0.4023	0.6542
M:N Relationship	0.6392	0.4563

Thus, we can positively answer the two research questions

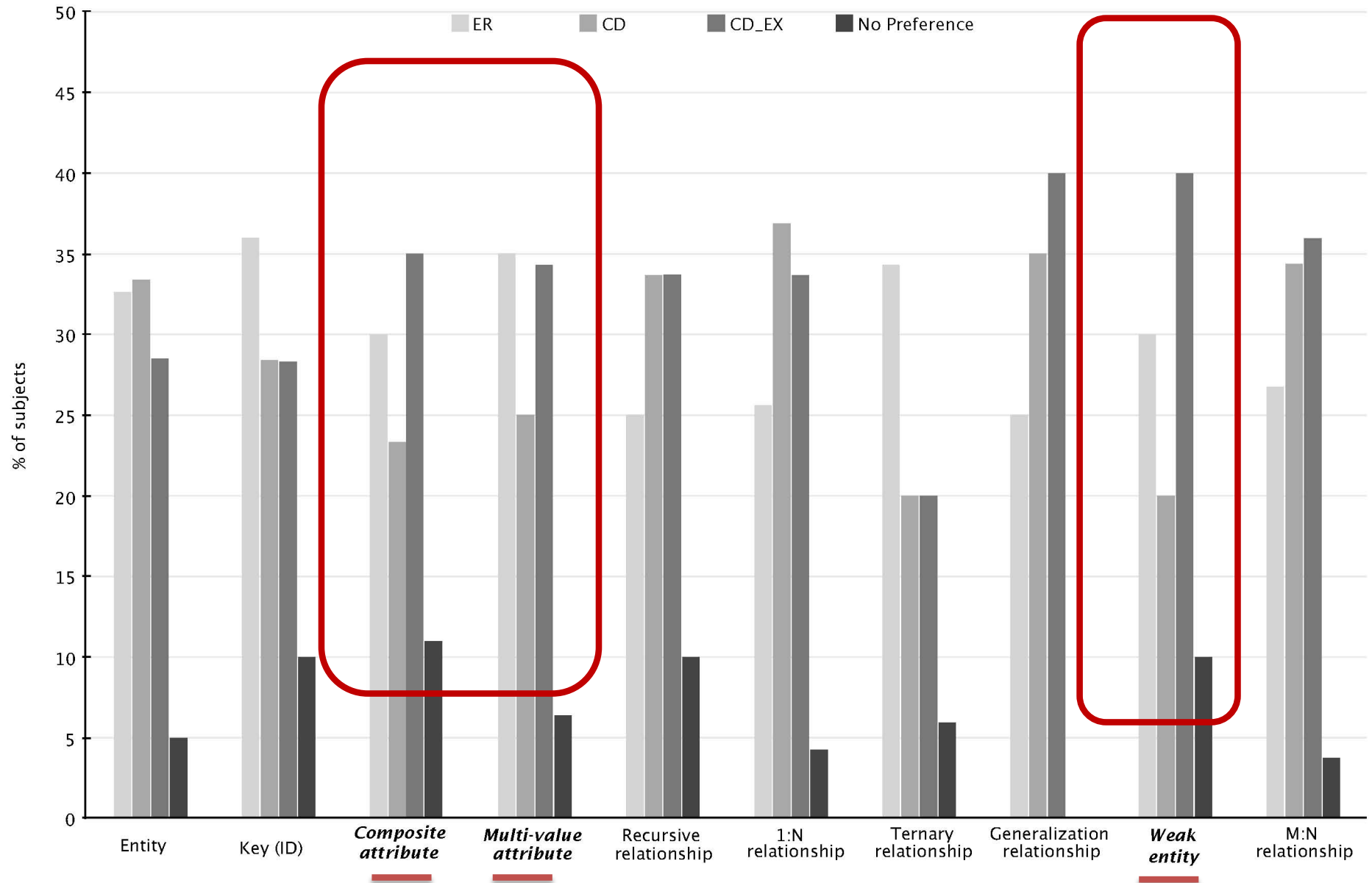
Results: Modification

- Preferences expressed by Bachelor students (IT)



Results: Modification (Cont.)

- Preferences expressed by Master students (IT)



Summary and Discussion

- **Weakness** of **CD** confirmed by all our experiments/replications
 - Qualitative analysis highlighted that for these three building blocks ER is more clear and explicit than the **CD**
 - **Possible reason:** **ER**, on the contrary of **CD**, employs customized graphical elements only used to represent specific concepts (e.g., the double circle for multivalued attributes)
- **CD** generally performs better than **ER** for the remaining seven building blocks, especially when supporting modification activities
- **CD_EX**, exploiting very **simply stereotypes** made representation of composite attribute, multi-value attribute, and weak entity clearer
 - *Strong improvements* in the performances of participants
 - qualitative analysis confirmed the *appreciation* of the employed stereotypes by participants

Summary and Discussion (cont.)

- Findings:
 - Using the *UML extension mechanisms* (in our case, stereotypes) it is possible to improve the effectiveness of UML diagrams in *specific environments* (in our case, data modeling) by defining customized concepts that are not so common in the standard usage of the UML notation
 - For *concepts widely used* in UML modeling (e.g., entity, 1:N relationship) CD did not show any failing with respect to ER
 - For *concepts not very frequently* used in UML modeling (e.g., weak entity, composite attribute), CD showed its *weak* side
 - ...

Future work

- Replications and further analysis
- The effect of different conceptual models upon reasoning in a database query writing task